

Developing
**Essential
Understanding**
of
Geometry

Grades 6–8

Developing
Essential Understanding
of

Geometry

for Teaching Mathematics in
Grades 6–8

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Foreword

Teaching mathematics in prekindergarten–grade 12 requires a special understanding of mathematics. Effective teachers of mathematics think about and beyond the content that they teach, seeking explanations and making connections to other topics, both inside and outside mathematics. Students meet curriculum and achievement expectations when they work with teachers who know what mathematics is important for each topic that they teach.

The National Council of Teachers of Mathematics (NCTM) presents the Essential Understanding Series in tandem with a call to focus the school mathematics curriculum in the spirit of *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics: A Quest for Coherence*, published in 2006, and *Focus in High School Mathematics: Reasoning and Sense Making*, released in 2009. The Essential Understanding books are a resource for individual teachers and groups of colleagues interested in engaging in mathematical thinking to enrich and extend their own knowledge of particular mathematics topics in ways that benefit their work with students. The topic of each book is an area of mathematics that is difficult for students to learn, challenging to teach, and critical for students' success as learners and in their future lives and careers.

Drawing on their experiences as teachers, researchers, and mathematicians, the authors have identified the big ideas that are at the heart of each book's topic. A set of essential understandings—mathematical points that capture the essence of the topic—fleshes out each big idea. Taken collectively, the big ideas and essential understandings give a view of a mathematics that is focused, connected, and useful to teachers. Links to topics that students encounter earlier and later in school mathematics and to instruction and assessment practices illustrate the relevance and importance of a teacher's essential understanding of mathematics.

On behalf of the Board of Directors, I offer sincere thanks and appreciation to everyone who has helped to make this series possible. I extend special thanks to Rose Mary Zbiek for her leadership as series editor. I join the Essential Understanding project team in welcoming you to these books and in wishing you many years of continued enjoyment of learning and teaching mathematics.

Henry Kepner
President, 2008–2010
National Council of Teachers of Mathematics

Preface

From prekindergarten through grade 12, the school mathematics curriculum includes important topics that are pivotal in students' development. Students who understand these ideas cross smoothly into new mathematical terrain and continue moving forward with assurance.

However, many of these topics have traditionally been challenging to teach as well as learn, and they often prove to be barriers rather than gateways to students' progress. Students who fail to get a solid grounding in them frequently lose momentum and struggle in subsequent work in mathematics and related disciplines.

The Essential Understanding Series identifies such topics at all levels. Teachers who engage students in these topics play critical roles in students' mathematical achievement. Each volume in the series invites teachers who aim to be not just proficient but outstanding in the classroom—teachers like you—to enrich their understanding of one or more of these topics to ensure students' continued development in mathematics.

How much do you need to know?

To teach these challenging topics effectively, you must draw on a mathematical understanding that is both broad and deep. The challenge is to know considerably more about the topic than you expect your students to know and learn.

Why does your knowledge need to be so extensive? Why must it go above and beyond what you need to teach and your students need to learn? The answer to this question has many parts.

To plan successful learning experiences, you need to understand different models and representations and, in some cases, emerging technologies as you evaluate curriculum materials and create lessons. As you choose and implement learning tasks, you need to know what to emphasize and why those ideas are mathematically important.

While engaging your students in lessons, you must anticipate their perplexities, help them avoid known pitfalls, and recognize and dispel misconceptions. You need to capitalize on unexpected classroom opportunities to make connections among mathematical ideas. If assessment shows that students have not understood the material adequately, you need to know how to address weaknesses that you have identified in their understanding. Your understanding must be sufficiently versatile to allow you to represent the mathematics in different ways to students who don't understand it the first time.

Big ideas and essential understandings are identified by icons in the books.



marks a big idea,
and



marks an essential understanding.

In addition, you need to know where the topic fits in the full span of the mathematics curriculum. You must understand where your students are coming from in their thinking and where they are heading mathematically in the months and years to come.

Accomplishing these tasks in mathematically sound ways is a tall order. A rich understanding of the mathematics supports the varied work of teaching as you guide your students and keep their learning on track.

How can the Essential Understanding Series help?

The Essential Understanding books offer you an opportunity to delve into the mathematics that you teach and reinforce your content knowledge. They do not include materials for you to use directly with your students, nor do they discuss classroom management, teaching styles, or assessment techniques. Instead, these books focus squarely on issues of mathematical content—the ideas and understanding that you must bring to your preparation, in-class instruction, one-on-one interactions with students, and assessment.

How do the authors approach the topics?

For each topic, the authors identify “big ideas” and “essential understandings.” The big ideas are mathematical statements of overarching concepts that are central to a mathematical topic and link numerous smaller mathematical ideas into coherent wholes. The books call the smaller, more concrete ideas that are associated with each big idea *essential understandings*. They capture aspects of the corresponding big idea and provide evidence of its richness.

The big ideas have tremendous value in mathematics. You can gain an appreciation of the power and worth of these densely packed statements through persistent work with the interrelated essential understandings. Grasping these multiple smaller concepts and through them gaining access to the big ideas can greatly increase your intellectual assets and classroom possibilities.

In your work with mathematical ideas in your role as a teacher, you have probably observed that the essential understandings are often at the heart of the understanding that you need for presenting one of these challenging topics to students. Knowing these ideas very well is critical because they are the mathematical pieces that connect to form each big idea.

How are the books organized?

Every book in the Essential Understanding Series has the same structure:

- The introduction gives an overview, explaining the reasons for the selection of the particular topic and highlighting some of the differences between what teachers and students need to know about it.

- Chapter 1 is the heart of the book, identifying and examining the big ideas and related essential understandings.
- Chapter 2 reconsiders the ideas discussed in chapter 1 in light of their connections with mathematical ideas within the grade band and with other mathematics that the students have encountered earlier or will encounter later in their study of mathematics.
- Chapter 3 wraps up the discussion by considering the challenges that students often face in grasping the necessary concepts related to the topic under discussion. It analyzes the development of their thinking and offers guidance for presenting ideas to them and assessing their understanding.

The discussion of big ideas and essential understandings in chapter 1 is interspersed with questions labeled “Reflect.” It is important to pause in your reading to think about these on your own or discuss them with your colleagues. By engaging with the material in this way, you can make the experience of reading the book participatory, interactive, and dynamic.

Reflect questions can also serve as topics of conversation among local groups of teachers or teachers connected electronically in school districts or even between states. Thus, the Reflect items can extend the possibilities for using the books as tools for formal or informal experiences for in-service and preservice teachers, individually or in groups, in or beyond college or university classes.

A new perspective

The Essential Understanding Series thus is intended to support you in gaining a deep and broad understanding of mathematics that can benefit your students in many ways. Considering connections between the mathematics under discussion and other mathematics that students encounter earlier and later in the curriculum gives the books unusual depth as well as insight into vertical articulation in school mathematics.

The series appears against the backdrop of *Principles and Standards for School Mathematics* (NCTM 2000), *Curriculum Focal Points for Prekindergarten through Grade 8 Mathematics: A Quest for Coherence* (NCTM 2006), *Focus in High School Mathematics: Reasoning and Sense Making* (NCTM 2009), and the Navigations Series (NCTM 2001–2009). The new books play an important role, supporting the work of these publications by offering content-based professional development.

The other publications, in turn, can flesh out and enrich the new books. After reading this book, for example, you might select hands-on, Standards-based activities from the Navigations books for your students to use to gain insights into the topics that the Essential Understanding books discuss. If you are teaching students

in prekindergarten through grade 8, you might apply your deeper understanding as you present material related to the three focal points that *Curriculum Focal Points* identifies for instruction at your students' level. Or if you are teaching students in grades 9–12, you might use your understanding to enrich the ways in which you can engage students in mathematical reasoning and sense making as presented in *Focus in High School Mathematics*.

An enriched understanding can give you a fresh perspective and infuse new energy into your teaching. We hope that the understanding that you acquire from reading the book will support your efforts as you help your students grasp the ideas that will ensure their mathematical success.

The authors wish to thank their reviewers—Karen Windham Wyatt, John K. Lannin, Gina M. Foletta, and Daniel Chazan—for their insightful comments on this volume. They are especially grateful for the contributions of David W. Henderson.

Introduction

This book focuses on ideas about geometry—ideas that you need to understand thoroughly and be able to use flexibly to be highly effective in your teaching of mathematics in grades 6–8. The book discusses many geometrical ideas that are common in middle school curricula, and it assumes that you have had a variety of mathematics experiences that have motivated you to delve into—and move beyond—the mathematics that you expect your students to learn.

The book is designed to engage you with these ideas, helping you to develop an understanding that will guide your planning and implementing of lessons and your assessing of your students' learning in ways that reflect the full complexity of geometry. A deep, rich understanding of geometric ideas will enable you to communicate their influence and scope to your students, showing them how these ideas permeate much of the mathematics that they have encountered—and will continue to encounter—throughout their school mathematics experiences.

The understanding of geometry that you gain from this focused study thus supports the vision of *Principles and Standards for School Mathematics* (NCTM 2000): “Imagine a classroom, a school, or a school district where all students have access to high-quality, engaging mathematics instruction” (p. 3). This vision depends on classroom teachers who “are continually growing as professionals” (p. 3) and routinely engage their students in meaningful experiences that help them learn mathematics with understanding.

Why Geometry?

Like the topics of all the volumes in NCTM's Essential Understanding Series, geometry constitutes a major area of school mathematics that is crucial for students to learn but challenging for teachers to teach. Students in grades 6–8 need to understand geometric ideas well if they are to succeed in these grades and in their subsequent mathematics experiences. Learners often struggle with ideas about geometry. Why are there so many formulas for area and volume, and why do some of them look alike, for example? What are some of the core images of geometry, and how might one work with them mentally, on paper, or on a screen? Why might one quadrilateral have no special name while another quadrilateral has several, such as *square*, *rectangle*, *rhombus*, and *parallelogram*, and why are these names useful? All these questions and many more make it essential for teachers of grades 6–8 to understand geometry extremely well themselves.

Your work as a teacher of mathematics in these grades calls for a solid understanding of the mathematics that you—and your school, your district, and your state curriculum—expect your students to learn about geometry. Your work also requires you to know how this mathematics relates to other mathematical ideas that your students will encounter in the lesson at hand, the current school year, and beyond. Rich mathematical understanding guides teachers' decisions in much of their work, such as choosing tasks for a lesson, posing questions, selecting materials, ordering topics and ideas over time, assessing the quality of students' work, and devising ways to challenge and support their thinking.

Understanding Geometry

Teachers teach mathematics because they want others to understand it in ways that will contribute to success and satisfaction in school, work, and life. Helping your students develop a robust and lasting understanding of geometry requires that you understand this mathematics deeply. But what does this mean?

It is easy to think that understanding an area of mathematics, such as geometry, means knowing certain facts, being able to solve particular types of problems, and mastering relevant vocabulary. For example, to teach geometry in the middle grades, you are expected to know the properties of several types of polygons. You are also expected to be skillful in determining the volume of various solids. Your mathematical vocabulary is assumed to include such terms as *quadrilateral*, *diameter*, *conjecture*, *vertex*, *angle*, and *midpoint*.

Obviously, facts, vocabulary, and techniques for solving certain types of problems are not all that you are expected to know about geometry. For example, in your ongoing work with students, you have undoubtedly discovered that you need not only to know common techniques for determining the area of polygons but also to be able to follow strategies that your students develop.

It is also easy to focus on a very long list of mathematical ideas that all teachers of mathematics in grades 6–8 are expected to know and teach about geometry. Curriculum developers often devise and publish such lists. However important the individual items might be, these lists cannot capture the essence of a rich understanding of the topic. Understanding geometry deeply requires you not only to know important mathematical ideas but also to recognize how these ideas relate to one another. Your understanding continues to grow with experience and as a result of opportunities to embrace new ideas and find new connections among familiar ones.

Furthermore, your understanding of geometry should transcend the content intended for your students. Some of the differences

between what you need to know and what you expect them to learn are easy to point out. For instance, your understanding of the topic should include a grasp of glide reflections as compositions of translations and reflections—mathematics that your students will encounter later but do not yet understand.

Other differences between the understanding that you need to have and the understanding that you expect your students to acquire are less obvious, but your experiences in the classroom have undoubtedly made you aware of them at some level. For example, how many times have you been grateful to have an understanding of geometry that enables you to recognize the merit in a student's unanticipated mathematical question or claim? How many other times have you wondered whether you could be missing such an opportunity or failing to use it to full advantage because of a gap in your knowledge?

As you have almost certainly discovered, knowing and being able to do familiar mathematics are not enough when you are in the classroom. You also need to be able to identify and justify or refute novel claims. These claims and justifications might draw on ideas or techniques that are beyond the mathematical experiences of your students and current curricular expectations for them. For example, you may need to be able to refute the often-asserted, erroneous claim that the base of a triangle always has to be a horizontal segment. Or you may need to explain to a student why volume is not always simply a product of length, width, and height.

Big Ideas and Essential Understandings

Thinking about the many particular ideas that are part of a rich understanding of geometry can be an overwhelming task. Articulating all of those mathematical ideas and their connections would require many books. To choose which ideas to include in this book, the authors considered a critical question: What is *essential* for teachers of mathematics in grades 6–8 to know about geometry to be effective in the classroom? To answer this question, the authors drew on a variety of resources, including personal experiences, the expertise of colleagues in mathematics and mathematics education, and the reactions of reviewers and professional development providers, as well as ideas from curricular materials and research on mathematics learning and teaching.

As a result, the mathematical content of this book focuses on essential ideas for teachers about geometry. In particular, chapter 1 is organized around four big ideas related to this important area of mathematics. Each big idea is supported by smaller, more specific mathematical ideas, which the book calls *essential understandings*.

Benefits for Teaching, Learning, and Assessing

Understanding geometry can help you implement the Teaching Principle enunciated in *Principles and Standards for School Mathematics*. This Principle sets a high standard for instruction: “Effective mathematics teaching requires understanding what students know and need to learn and then challenging and supporting them to learn it well” (NCTM 2000, p. 16). As in teaching about other critical topics in mathematics, teaching about geometry requires knowledge that goes “beyond what most teachers experience in standard preservice mathematics courses” (p. 17).

Chapter 1 comes into play at this point, offering an overview of the topic that is intended to be more focused and comprehensive than many discussions that you are likely to have encountered. This chapter enumerates, expands on, and gives examples of the big ideas and essential understandings related to geometry, with the goal of supplementing or reinforcing your understanding. Thus, chapter 1 aims to prepare you to implement the Teaching Principle fully as you provide the support and challenge that your students need for robust learning about geometry.

Consolidating your understanding in this way also prepares you to implement the Learning Principle outlined in *Principles and Standards*: “Students must learn mathematics with understanding, actively building new knowledge from experience and prior knowledge” (NCTM 2000, p. 20). To support your efforts to help your students learn about the concepts in this way, chapter 2 builds on the understanding of geometry that chapter 1 communicates by pointing out specific ways in which the big ideas and essential understandings connect with mathematics that students typically encounter earlier or later in school. This chapter supports the Learning Principle by emphasizing longitudinal connections in students’ learning about geometry. For example, as their mathematical experiences expand, students gradually develop an understanding that the invariance of relationships among similar right triangles underlies trigonometric functions.

The understanding that chapters 1 and 2 convey can strengthen another critical area of teaching. Chapter 3 addresses this area, building on the first two chapters to show how an understanding of geometry can help you select and develop appropriate tasks, techniques, and tools for assessing your students’ understanding of geometry. An ownership of the big ideas and essential understandings related to geometry, reinforced by an understanding of students’ past and future experiences with related ideas, can help you ensure that assessment in your classroom supports the learning of significant mathematics.

Such assessment satisfies the first requirement of the Assessment Principle set out in *Principles and Standards*: “Assessment should support the learning of important mathematics and furnish useful information to both teachers and students” (NCTM 2000, p. 22). An understanding of geometry can also help you satisfy the second requirement of the Assessment Principle, by enabling you to develop assessment tasks that give you specific information about what your students are thinking and what they understand.

For example, chapter 3 explores some specific mathematical questions in this context: How would students construct a polygon that has all angles congruent but non-congruent sides? Would they recognize non-square rectangles as related to the quadrilateral case? Would they be able to explain why no such triangle exists? Tasks and questions like these can provide insight into how your students understand equilateral and equiangular shapes and how they reason geometrically across situations that offer a variety of familiar results, impossible constraints, or infinitely many solutions.

Ready to Begin

This introduction has painted the background, preparing you for the big ideas and associated essential understandings related to geometry that you will encounter and explore in chapter 1. Reading the chapters in the order in which they appear can be a useful way to approach the book. Read chapter 1 in more than one sitting, allowing time for reflection. Take time also to use the wide range of tools recommended, including dynamic geometry software. Absorb the ideas—both big ideas and essential understandings—related to geometry. Appreciate the connections among these ideas. Carry your newfound or reinforced understanding to chapter 2, which guides you in seeing how the ideas related to geometry are connected to the mathematics that your students have encountered earlier or will encounter later in school. Then read about teaching, learning, and assessment issues in chapter 3.

Alternatively, you may want to take a look at chapter 3 before engaging with the mathematical ideas in chapters 1 and 2. Having the challenges of teaching, learning, and assessment issues clearly in mind, along with possible approaches to them, can provide you with a different perspective on the material in the earlier chapters.

No matter how you read the book, let it serve as a tool to expand your understanding, application, and enjoyment of geometry.

Geometry: The Big Ideas and Essential Understandings

Geometry, as an integral area of mathematics, often loses its way in the middle school grades. An increasing emphasis on early algebra, as well as on working with variables in deriving general formulas, can make what is essentially geometric about a situation fade into the background or even disappear. For this reason, we have opted to start this chapter with an assertion about the centrality of geometry, both as an area of study in its own right and as a means of providing insight and understanding for other areas of mathematics. We then turn to the importance of imagery, the nature of geometric figures, and the influence of tools and geometric activities.

Four big ideas and several smaller, more specific essential understandings provide the structure of this chapter. The big ideas and all the associated understandings are identified as a group below to give you a quick overview and for your convenience in referring back to them later. Read through them now, but do not think that you must absorb them fully at this point. The chapter will discuss each one in turn in detail.

Big Idea 1. Behind every measurement formula lies a geometric result.



Essential Understanding 1a. Decomposing and rearranging provide a geometric way of both *seeing that* a measurement formula is the right one and *seeing why* it is the right one.

Essential Understanding 1b. In addition to decomposing and rearranging, shearing provides another geometric way of both *seeing that* a measurement formula is the right one and *seeing why* it is the right one.





Big Idea 2. Geometric thinking involves developing, attending to, and learning how to work with imagery.

Essential Understanding 2a. Geometric images provide the content in relation to which properties can be noticed, definitions can be made, and invariances can be discerned.



Essential Understanding 2b. Symmetry provides a powerful way of working geometrically.

Essential Understanding 2c. Geometric awareness develops through practice in visualizing, diagramming, and constructing.



Big Idea 3. A geometric object is a mental object that, when constructed, carries with it traces of the tool or tools by which it was constructed.

Essential Understanding 3a. Tools provide new sources of imagery as well as specific ways of thinking about geometric objects and processes.



Essential Understanding 3b. Geometric thinking turns tools into objects, and in geometry the process of turning an action undertaken with a tool into an object happens over and over again.



Big Idea 4. Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.

Essential Understanding 4a. Naming is not just about nomenclature: it draws attention to properties and objects of geometric interest.



Essential Understanding 4b. Definition can both generate and reflect structure: definitions are often dependent on a specific classification.

Essential Understanding 4c. Conjectures can emerge out of a problem-posing process that generates claims that need to be justified.

The Tangled Relationship between Measurement Formulas and Geometry: Big Idea 1

Big Idea 1. *Behind every measurement formula lies a geometric result.*

Measurement (and the use of formulas to produce numerical answers to measurement tasks) can seem to be the primary focus of geometry in the middle grades. For some people, however—including the authors on most days—most measurement lies outside geometry—despite the “-metry” in the word *geometry*!

The whole is the sum of its parts

Essential Understanding 1a. *Decomposing and rearranging provide a geometric way of both seeing that a measurement formula is the right one and seeing why it is the right one.*

When we are comparing the areas of different shapes, our first impulse is often to think numerically: do the measurements, in some square units of measure, yield the same number? Measurements may spring to mind quickly, because we have powerful formulas for determining them, such as the one for the area of a triangle.

In the ancient Greek approach to such a question, numbers would be completely absent. The area of a triangle, say, would be compared with the area of another geometric shape. But which shape? That would depend on the context. One very powerful means through which such a comparison might occur involves decomposition and rearrangement. Reflect 1.1 involves rearrangement with tangram pieces.

Reflect 1.1

Rearrange the seven pieces of a tangram into different designs. Start with a square. Then make other shapes. Convince yourself that your shapes all have the same area.

The shapes in figure 1.1 look quite different, and your first impression might be that some of them cannot possibly have the same area. But each one is composed of the exact same seven tangram pieces, and this means that they must be equal in area. A key fact about area is that it is *additive*: the area of the composition of two figures is the sum of their areas. This assures us that each one of the shapes in figure 1.1 has to have the same area as every other one. Thinking about area in this way focuses attention on comparing



areas through decomposition and rearrangement, rather than by calculating numerical values. Reflect 1.2 invites you to compare areas of triangles and quadrilaterals.

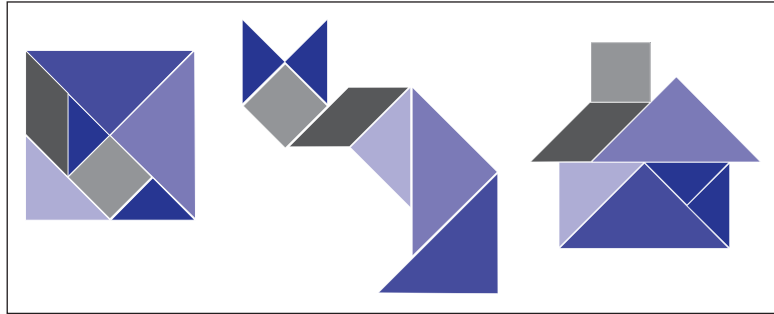


Fig 1.1. Shapes made with tangram pieces

Reflect 1.2

Draw a triangle. Now draw a rectangle that has the same area as your triangle. Can you draw a different quadrilateral (a non-rectangular one, perhaps) that also has the same area as the starting triangle? What about a polygon with more than four sides?

We could do the very same kind of work with triangles as with the tangram shapes, although slightly less playfully, to say something about their areas. In figure 1.2, an arbitrary triangle T is first decomposed into two parts N and M by the dotted line representing an altitude of the triangle. A second dotted line that is parallel to the base and passes through the midpoint of the altitude leads to a further decomposition of the original triangle into four parts, which we have labeled A , B , C , and D . By rearranging C and D (by rotating them 180 degrees about the points of intersection of the halfway line and the respective sides of the triangle), we transform triangle T into rectangle R . The triangle in this figure is acute. Any triangle, whether acute or obtuse, always has a longest side (though in the case of an isosceles or equilateral triangle this side is not unique). If we consider a rotation of an obtuse triangle—a change that does not alter area—we can think of that obtuse triangle as oriented as in figure 1.2.

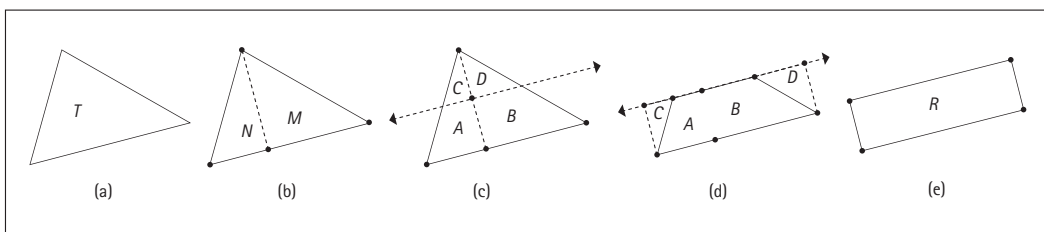


Fig. 1.2. Decomposing and rearranging a triangle into a rectangle

The diagrams in figure 1.2 tell a compelling story. But we can make the story even better by explaining why shape R is a rectangle. Figure 1.3a zooms in on just the left side of the configuration in figure 1.2c. Because the line passing through the midpoint of the altitude is parallel to the base, it cuts the altitude at right angles. The right angle marked in figure 1.3a is at one “corner” of R . How do we know that the other corner of R (the lower left vertex in figure 1.3b) is also a right angle? Because we are doing a half-turn, we know that the half-altitude of C (indicated by the double hash marks on one leg of triangle C), which was perpendicular to the base of T to start with, will still be perpendicular to the base of T , and parallel to the original altitude. A similar argument works for triangle D in relation to B .

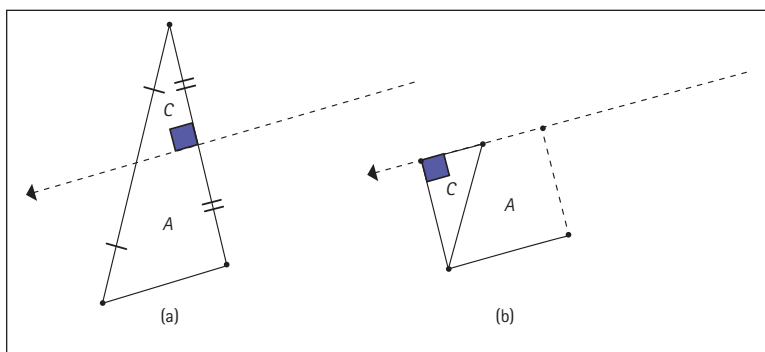


Fig. 1.3. Rotating C to create the left edge of the rectangle R

In this way, we can say that the area of T is the same as the area of R , so we have found the area of a triangle in terms of the area of an equivalent rectangle. We note that the length of the rectangle is the same as the base of the original triangle and that its width is exactly one-half of the height of the triangle. In this manner, we can compare the area of the triangle with the area of a rectangle, leading to the well-known formula for the area of a triangle: $A = \frac{1}{2}bh$.

If T is an obtuse triangle, we can still undertake an identical process, provided that we choose the longest side as the base of the altitude (see the alternative shown in fig. 1.4). However, we could also create a comparable dissection argument for this obtuse triangle by choosing a different altitude from the one shown in figure 1.4. We would still create the rectangle of base times half-height. The paired triangles in figure 1.5d and 1.5e are the same in that one is a rotation through one-half the angle of rotation of the other (this claim still needs a formal proof), but the sequence of decomposition and rearrangement in figure 1.5 explains *why* the same formula holds even for an obtuse angle triangle (though notice that the *base* and the *height* are different here—why does this not seem to matter in the formula?).

The word *height* sometimes means the same thing as *altitude* and sometimes is usefully distinguished, referring to the *measure* of the altitude. Furthermore, *height* is sometimes interpreted as vertical height, whereas an altitude can lie in any direction but is always perpendicular to a corresponding base.

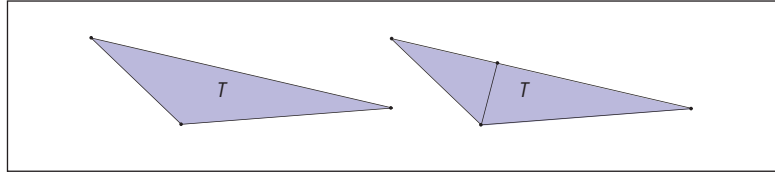


Fig. 1.4. Picking the height that corresponds to the longest side of the triangle

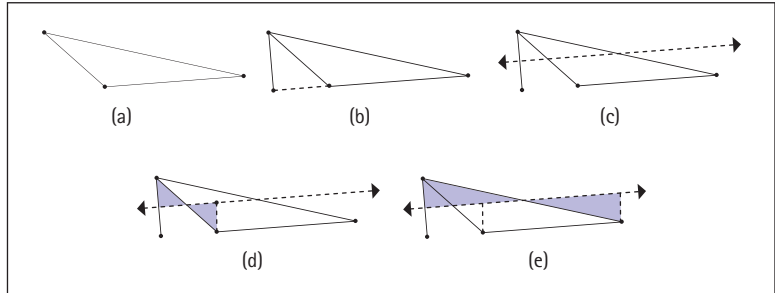


Fig. 1.5. The case of the decomposition of an obtuse-angled triangle, showing area is base times half-height

Through a different process of decomposition and rearrangement, we can also decompose the triangle T in figure 1.2 into two different triangles, N and M (see fig. 1.6). But this time, we draw the dotted line parallel to the base and passing through the third vertex, so that it intersects the two lines perpendicular to the base and passes through its endpoints to produce a rectangle containing copies of triangles N and M . Consequently, this rectangle will have twice the area of T , and this means that it can be split into two congruent rectangles, each having the same area as T . This immediately shows that the area of T is one-half of the rectangle's base times its height.

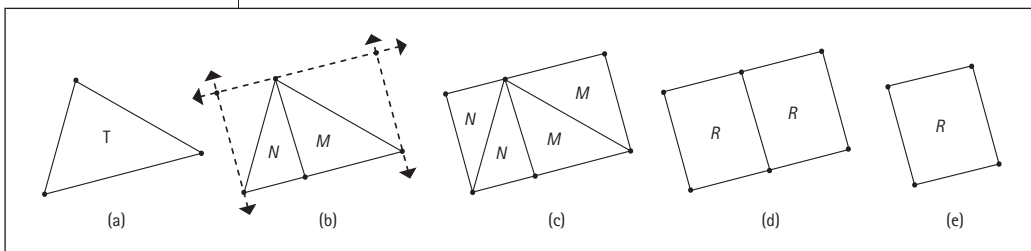


Fig. 1.6. Decomposing and rearranging a triangle into a different rectangle

Further, if the split occurs along the line perpendicular to the base and through the third vertex of T , then we get two copies of rectangle R , each of whose length is equal to half the base and each of whose width is equal to the height of the triangle, leading to the formula $A = (\frac{1}{2}b)h$. As was the case with the previous approach to decomposition, an obtuse triangle can be handled by taking the

longest side to be the base. We offer you the challenge of creating a set of diagrams showing this decomposition approach for an obtuse triangle, in the manner of figure 1.6.

We chose to compare triangle T with rectangles in both cases, and for a good reason. Using our principle of decomposition and rearrangement, we might also have chosen to compare T with a variety of other shapes, as shown in figure 1.7. The first one (fig. 1.7a) shows a different way of rearranging C and D from the one shown in figure 1.2c, producing a heptagon instead of a rectangle. Figure 1.7b shows a division of the large rectangle (composed of two copies of R) into two right triangles—thus showing the area of T by comparing it with another triangle. Figures 1.7c and 1.7d show alternate ways of halving the rectangle—into trapezoids and hexagons, respectively. The fact that all these shaded shapes have the same area, which is the area of T —as well as the area of the rectangle in figure 1.2e and the rectangle in figure 1.6e—is a consequence of the decomposing and rearranging.

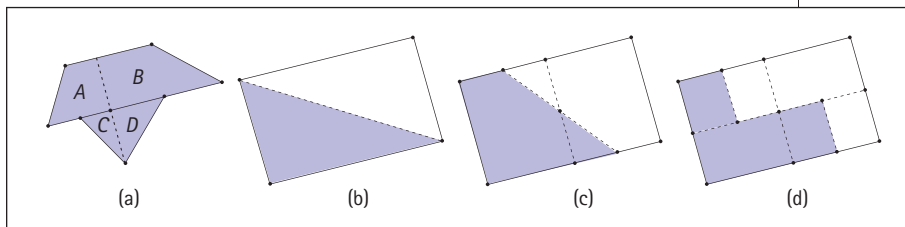


Fig. 1.7. Decomposing and rearranging a triangle into other shapes

Let's call the rectangle in figure 1.2e R_1 and the rectangle in figure 1.6e R_2 . In comparing T with both R_1 and R_2 by using the triangle area formula, we notice the rather surprising fact that although the two rectangles R_1 and R_2 have distinct shapes, they have the same area (see fig. 1.8). This, of course, is something that the formula itself asserts, since algebraically, $(\frac{1}{2}b)h = b(\frac{1}{2}h)$, where b and h are now seen as numbers—length measures of sides. A third algebraically equivalent form is $\frac{1}{2}(bh)$. This formula has a geometric interpretation as one-half the area of a rectangle, as suggested by figures 1.7b, 1.7c, and 1.7d. However, bear in mind that the algebraic results arose from quite *different* ways of *seeing* the decomposition or rearrangement of a triangle geometrically. Algebra frequently masks distinctions that are clear and distinct in the geometry.

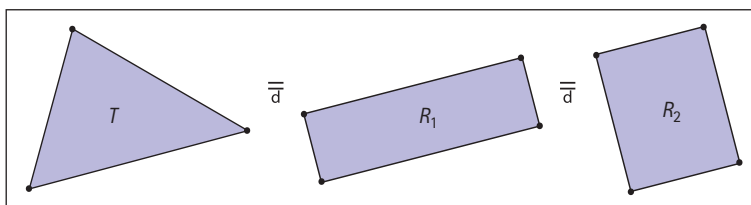


Fig. 1.8. Triangle T and rectangles R_1 and R_2 have the same area

The symbol $\equiv$ is sometimes used to indicate that two geometric figures are "equivalent by dissection," which means, in particular, that they have the same area.

This is not the only sense in which the algebra asserts a sameness that the geometry does not so obviously show. In figure 1.2, one side of triangle T was chosen as the base, but any of the three sides *could* have played that role, with one of three different, corresponding segments then being called the *height*. Working with this first approach to decomposition and rearrangement, we obtain rectangles Q_1 , R_1 , and S_1 , all different from one another but each having the same area as triangle T (see fig. 1.9a). Working with our second approach, we obtain the triangles R_2 , Q_2 , and S_2 (see fig. 1.9b). The triangle area formula is powerful because it works for any of the pairs of bases and corresponding heights that we choose and because it asserts that these six seemingly different rectangles are all equal in area to one another—and to T . This result is why we can get away with being casual about the form of the area formula for a triangle, asserting simply that the area is one-half base times height.

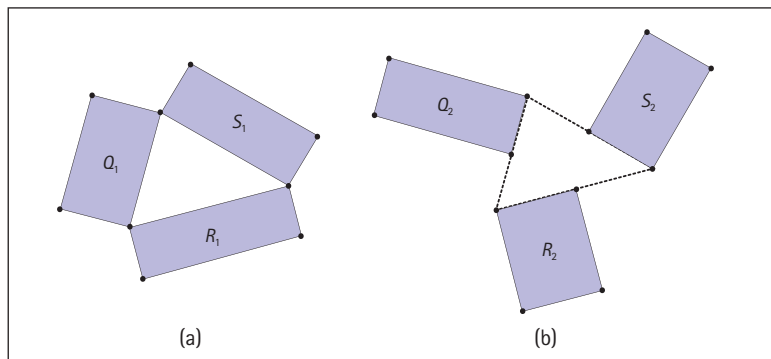


Fig. 1.9. Decomposing and rearranging a triangle into rectangles with different bases and corresponding heights

Figure 1.9 shows six rectangles with the same area as one another and as triangle T . Reflect 1.3 asks you to consider parallelograms that have exactly twice the area of a triangle.

Reflect 1.3

Given a triangle T , how many different parallelograms can you find that are exactly twice the area of T ?

In textbooks, the area of a triangle is often compared with the area of a parallelogram, since any parallelogram can be divided into two congruent triangles just by drawing in one of its diagonals. Equivalently, any triangle can be rotated around the midpoint of one of its sides to produce another triangle that can be composed with the original triangle to form a parallelogram. So the area of any triangle can easily be compared with the area of a parallelogram. Dwelling on this idea by exploring the different ways in which one

can produce a parallelogram out of a triangle helps to emphasize the idea of area as a *comparison between two shapes* rather than as a number. A premature shift to the algebraic formula can get in the way of developing the geometric insights that underlie any measurement formula.

In the Expectations for grades 6–8 of NCTM's Geometry Standard (NCTM 2000), we find, "All students should ... create and critique inductive and deductive arguments concerning geometric ideas and relationships such as congruence, similarity, and the Pythagorean relationship" (p. 232). As our final example of decomposing and rearranging in relation to Essential Understanding 1a, we look at the most well-known theorem of school mathematics—namely, the Pythagorean theorem. Here, and again at the end of the next section in connection with Essential Understanding 1b, we explore ways of seeing *why* this theorem is true. Our exploration begins with Reflect 1.4.

Essential Understanding 1b

In addition to decomposing and rearranging, shearing provides another geometric way of both seeing that a measurement formula is the right one and seeing why it is the right one.

Reflect 1.4

Below are two slightly different ways in which some people state the Pythagorean theorem:

1. For any right triangle in the plane, the square on the hypotenuse is always equal to the sum of the squares on the other two sides.
2. For any right triangle in the plane, the square of the hypotenuse is always equal to the sum of the squares of the other two sides.

What difference do you notice and what mathematical significance, if any, does this difference have?

As you might expect from our inclusion of the Pythagorean theorem in our discussion of Big Idea 1, we are interested in this theorem as a theorem of geometry, as opposed to its common algebraic interpretation. Further, here we are interested in attempting to see *why* the theorem is true, by using decomposition and rearrangement. The difference between the two statements in Reflect 1.4 is the choice of preposition (*on* versus *of*) with regard to the relationship between the square and the sides of the triangle. In talking about the square *on* a side of the triangle, we emphasize that a geometric object is placed in relation to another geometric object, as shown in figure 1.10.

The square *of* the hypotenuse refers to the numerical operation of squaring a number (representing the *length*

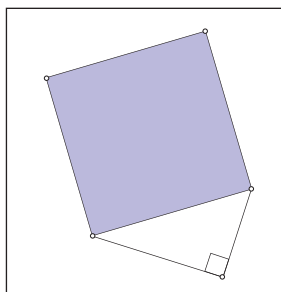


Fig. 1.10. The shaded square is "the square on the hypotenuse."

Big Idea 1



Behind every measurement formula lies a geometric result.

of the hypotenuse). Choosing this preposition is enough by itself to put us into an arithmetic or algebraic frame of mind, in which the Pythagorean relationship is conceived as an equality of numbers rather than as a sameness of areas. If we are thinking about sameness of areas, decomposing and rearranging the square on the hypotenuse so that it fits exactly into the other two squares is a plausible avenue for an informal (leading to a formal) proof. The question then becomes, “How should the square on the hypotenuse be decomposed? If we were going to cut the square into pieces, how do we know where to cut?”

One broad heuristic is to overlap the relevant interior regions as much as possible and then concentrate on what is left. Figure 1.11 shows first one, then a second, reflection of squares across a side of the original right triangle (transforming $ACDE$ to $ACD'E'$ and, subsequently, $BCGF$ to $BCG'F'$). The third construction involves dropping a perpendicular from D' onto CH .

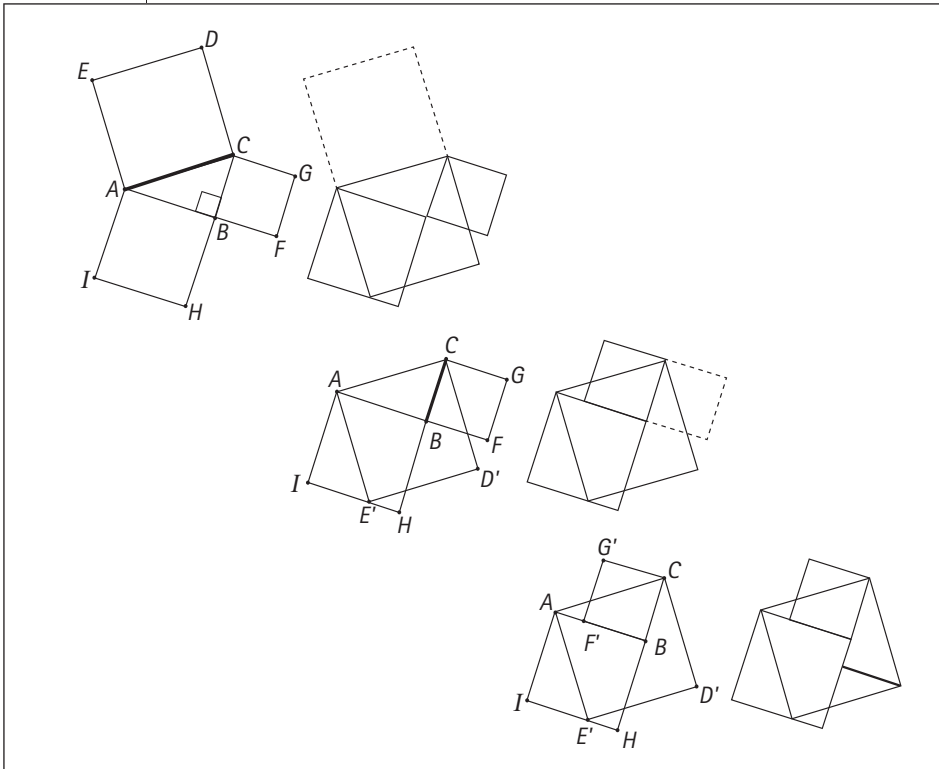


Fig. 1.11. A sequence of three constructions involved in a dissection proof of the Pythagorean theorem

Figure 1.12 shows how the five polygons into which the square on the hypotenuse has been cut in figure 1.11 can then be rearranged to fit exactly onto the two squares on the other sides of the right

triangle. This proof is still informal, in the sense that we have not yet shown the exactness with which these pieces fit together, nor do we, for instance, yet know for certain that one vertex of the square on the hypotenuse (point E in fig. 1.11) will always fall precisely onto the side of one of the other squares (E'). However, other than this uncertainty, the precise points at which the cuts are to be made are determined by the constructions.

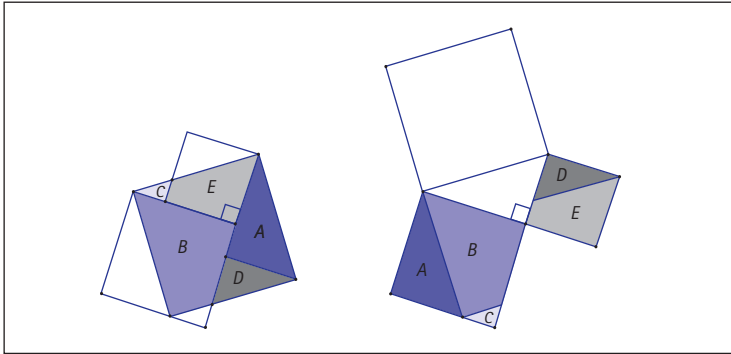


Fig. 1.12. Rearranging the pieces of the square on the hypotenuse to fit into the two squares on the sides of the right triangle

Further empirical as well as heuristic evidence can be found by constructing these figures in a dynamic geometry environment (DGE), such as The Geometer's Sketchpad or Cabri Geometry, and, for example, dragging the vertices of the original right triangle to see how the configuration and its decomposition hold together. In particular, looking at special cases, as when the right triangle is isosceles or when one leg is very small, can help provide a continuous sense of cases.

In our discussion of Essential Understanding 1a, we have focused on area formulas relating to triangles, but similar points underlying measurement formulas hold for all such results taught in school geometry, both in the plane and in three dimensions. For three-dimensional shapes, decomposing and rearranging solids can accomplish the work of making the formulas geometrically meaningful, though this can prove much more challenging.

To start to see the volume of a square-based pyramid, for instance, one can begin with a simple decomposition of a cube into three identical and specific square-based pyramids, as shown in figure 1.13. From this decomposition, one can see that the volume of each pyramid is exactly one-third of the volume of the original cube. For this decomposition, we have had to use a very particular type of square-based pyramid: it has the same height as the sides of the square base and it is right-angled at one corner. Both these attributes are required so that parts of these pyramids match the outside faces and corners of the starting cube. Consequently, we have

shown this relationship only for this very specific type of pyramid. In the discussion that follows of Essential Understanding 1b, we will work toward showing that the same volume formula holds for a general pyramid.

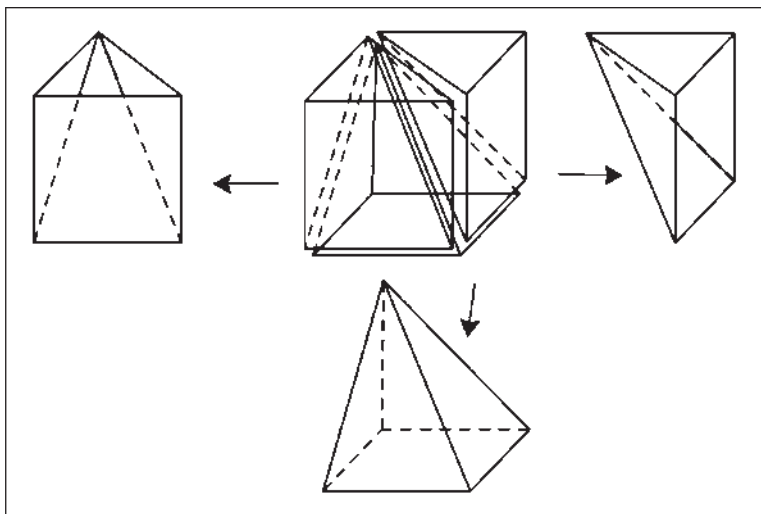


Fig. 1.13. Decomposition of a cube into three identical square-based pyramids

Continuous decomposition and infinite rearrangement



Essential Understanding 1b. *In addition to decomposing and re-arranging, shearing provides another geometric way of both seeing that a measurement formula is the right one and seeing why it is the right one.*

Shearing is a geometric transformation that makes a continuous and systematic change to a figure in such a way that it does not alter the figure's area. With Reflect 1.5, we begin our exploration of shearing by focusing on how we might know that two triangles have the same area.

Reflect 1.5

Draw five different triangles that have the same area. How do you know that their areas are the same?

Working from the formula, you might have generated pairs of numbers that have the same product and then used these numbers to build the sides of different triangles. Or perhaps you worked more geometrically by starting with rectangles that have the same area

and then halving them to produce triangles. Another, more continuous approach draws on Cavalieri's principle, which involves *shearing*, a particular sort of transformation.

For two dimensions, Cavalieri expressed his result as follows: "If between the same parallels any two plane figures are constructed, and if in them, any straight lines being drawn equidistant from the parallels, the included portions of any one of these lines are equal, the plane figures are also equal to one another" (Cavalieri 1635; in Proulx and Pimm 2008, p. 17). For a triangle, for example, Cavalieri's principle holds that the area does not change when the third vertex varies along a line parallel to the line through the other two vertices. So, in figure 1.14, for example, triangles CBD , CBA , CBE , and CBF all have the same area. The triangles may all look different—indeed, if shown these triangles separately, most people would believe that triangle CBA has a greater area than the others and that triangle CBF has a smaller one than the others—but they all in fact have the same base, BC , and the same height. Dragging point A in a dynamic geometry environment can help illustrate the property of shearing. By measuring the lengths of the sides AB and AC , as well as the area of ABC , one can see the lengths of the sides changing while the area remains invariant.

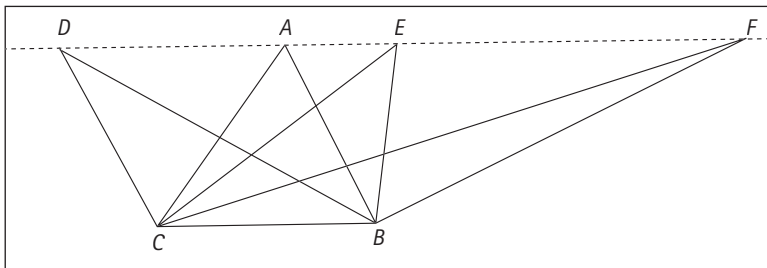


Fig. 1.14. Shearing a triangle to create other triangles all with the same area

Cavalieri's principle also works for shapes other than triangles. In particular, consider the quadrilaterals shown in figure 1.15. The rectangle on the left is sheared to become a parallelogram that is slightly slanted and then another, which is even more slanted. By Cavalieri's principle, all three of these shapes have the same area. The relationship between a parallelogram and a rectangle comes gratis—that is, it provides an alternative way of comparing areas without the cost of finding and working with numerical values. Note that repeated applications of Cavalieri's principle to a decomposed and then reassembled rectangle allow us to claim that the two figures shown in figure 1.16 have the same area.

We now return to look at Pythagoras's theorem as it might appear through the eyes of Cavalieri. What would a shearing proof

Bonaventura Cavalieri (1598–1647) was an Italian mathematician who was involved in the early development of the geometric calculus and corresponded extensively with Galileo.

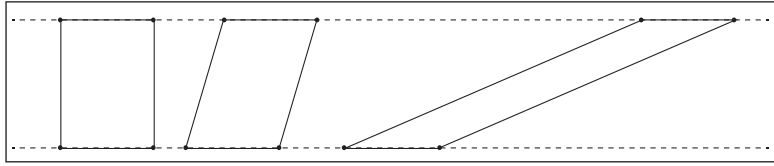


Fig. 1.15. Shearing a rectangle to produce parallelograms with the same area

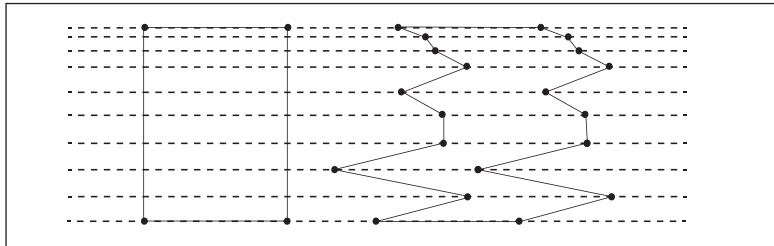


Fig. 1.16. Decomposing, shearing, and rearranging to obtain polygons of equal area

of this result look like? Figures 1.17 and 1.18 show a sequence of four shears, two applied to each of the two smaller squares. With any shear, you need to know the direction (what line you are moving parallel to) and the destination of the shearing motion. Figure 1.17b shows a line passing through the vertex of the right angle and perpendicular to the hypotenuse. This line suggests the destination of the first two shears. In figure 1.17c, the shearing is parallel to the horizontal leg of the right triangle, resulting in a transforming of the square on this leg into a parallelogram with the same area. Last, a similar shear is carried out with the square on the vertical leg of the triangle: the destination is the same, but this time the shearing is parallel to the other leg of the triangle (see fig. 1.17d).

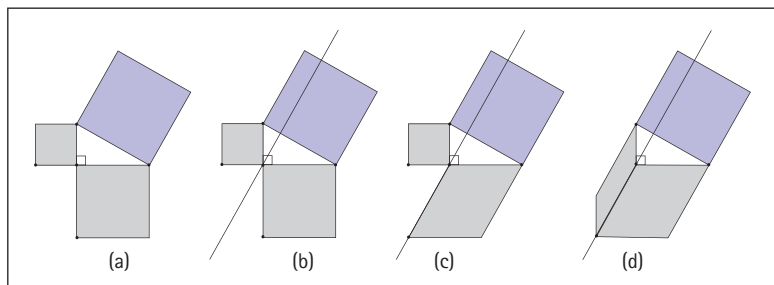


Fig. 1.17. Shearing the two squares on the legs of the right angle triangle

The final two shears in figure 1.18 are parallel to the line constructed at the beginning (as in fig. 1.17b), and the destination is suggested by the hypotenuse. They both transform parallelograms into rectangles with the same areas as before. The composition of

the two rectangles thus produced is a square—one that is identical to the square on the hypotenuse. This is still an informal proof, in the sense that we have not yet shown that the two rectangles form a square or that the two squares in figure 1.18c have exactly the same area.

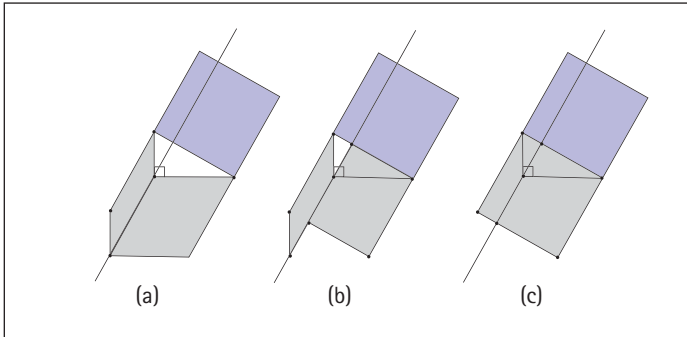


Fig. 1.18. Shearing the parallelograms into rectangles

Cavalieri's principle is also applicable to volumes. For three dimensions, Cavalieri expressed his result as follows: "If between the same parallel planes any solid figures are constructed, and if in them, any planes being drawn equidistant from the parallel planes, the included plane figures out of any one of the planes so drawn are equal, the solid figures are likewise equal to one another" (Cavalieri 1635; Proulx and Pimm 2008, pp. 17–18). This principle can be expressed as follows: two solids have the same volume if they lie between two parallel planes (hence have the same height), and any section made by a plane parallel to the plane of the common base of both solids creates cross sections with the same area. Figure 1.19 illustrates the principle for a rectangular prism. The dynamic dragging of a DGE can illustrate both the continuity and range of same-volume prisms.

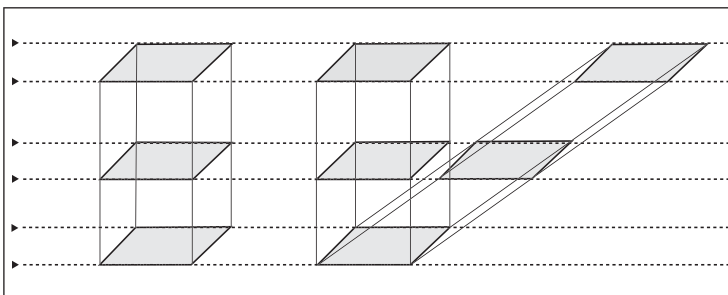


Fig. 1.19. Shearing a rectangular prism

A similar experiment can be performed with a stack of pennies. Arranging the pennies in a perfect cylinder would produce a solid with the same volume as the solid produced by tilting the cylinder or even by clumsily stacking the pennies one on top of another. For

both the prism and the cylinder, the shearing approach to thinking about volume draws attention to the way in which the solid is composed of identical shapes stacked one on top of another. If one knows the area of the base (B), then the volume is simply that area times the height of the stack. The volume of a rectangular prism can thus be seen to be base area times height (and seen more directly than as given in the conventional formula $V = l \times w \times h$). For more on using Cavalieri in working with areas and volumes in school geometry, see Proulx and Pimm (2008).

Cavalieri's principle reveals an even more general relationship: once we know the volume of a right square-based pyramid (or one with any shape of base), we can be sure that a pyramid that has the same height and base but has been sheared will have the same volume (see fig. 1.20). The attentive reader will have noticed that we have not yet talked about how to find the volume of a square-based pyramid whose height is different from the side of the base square.

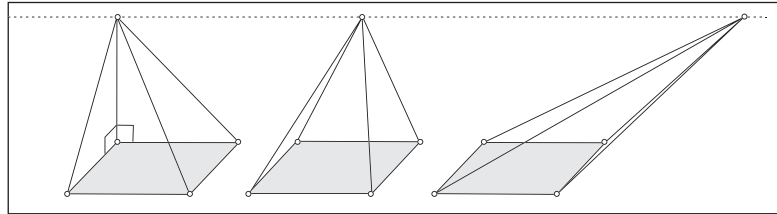


Fig. 1.20. Three pyramids with the same volume

It can be shown that the volume of a square-based pyramid whose height is not the same as the length of its base is also one-third the volume of the square prism with the same height. The process involves cutting the prism into six triangle-based prisms and rearranging each pair of these into a square-based prism. Although we could pursue that path, we would like to focus here on the somewhat easier case of determining the relationship between the triangle-based pyramid and the corresponding triangular prism, an argument that involves decomposing and rearranging as well as shearing.

We start with a triangular prism and decompose it into three triangular pyramids, as shown in figure 1.21a, 1.21b, and 1.21c. A 180-degree rotation about the midpoint of the common edge lying on the face ADC of the triangular prism would transform one pyramid into another and thus confirm that pyramids $ABCD$ and $FEDA$ have the same volume.

The third pyramid (see fig. 1.21c) has a different shape, but if we take ECB as its base and shear point D down toward A , we will not change its volume (see fig. 1.22). This shearing will eventually result in a pyramid that has the same base and height as the pyramids shown in figure 1.21a and 1.21b, therefore demonstrating to us that the three pyramids in figure 1.21 have the same volume.

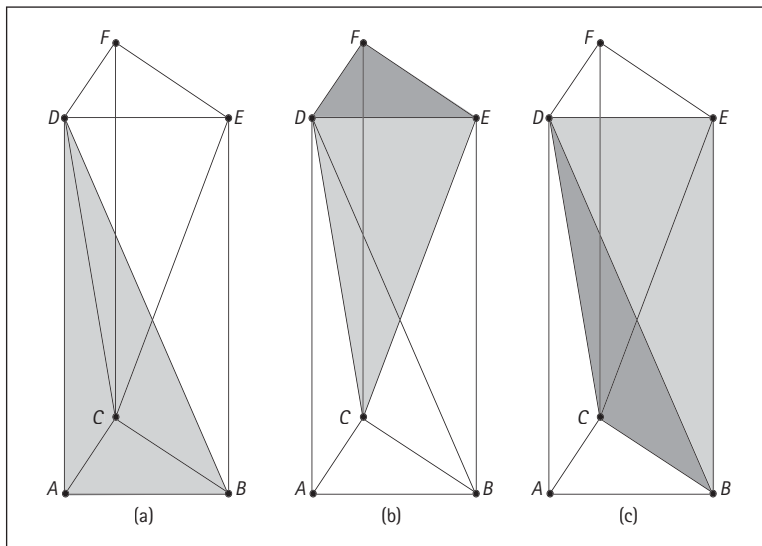


Fig. 1.21. Decomposing the triangular prism into three triangular pyramids

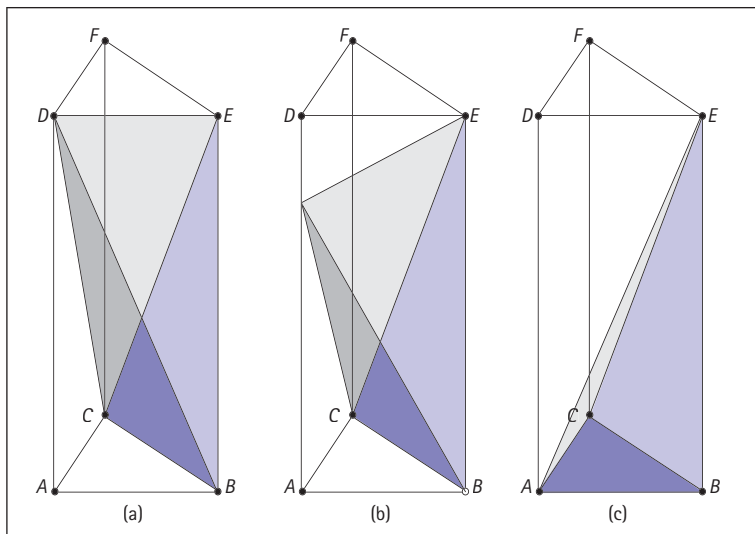


Fig. 1.22. Shearing of a triangular prism

This process of decomposition and shearing shows that a non-tilted triangular prism has the same volume as the composition of three identical triangular pyramids, each with the same height and base as the prism. If we had started with a tilted prism, a simple shear would be enough to bring us back to the case of the right triangular prism that we have argued for here.

Both shearing and decomposing allow us to approach questions of area and volume by comparing one shape with another through the use of transformation principles that preserve area and volume. The shift to number-based measurement, and to formulas, involves comparing attributes of shapes to specifically chosen units (seen as

numbers rather than geometric figures). So, for volume, one measures in cubic units (cubic inches, cubic centimeters, cubic whatever). Similarly, for area, one measures in square units (square inches, square centimeters, square whatever).

But this need not be so. To provide a brief coda to this section, we note that when working, for example, with the areas of the commonly studied quadrilaterals, a rectangle that is not necessarily a square seems like a very suitable candidate with which to compare other shapes. For example, the area of a triangle is half the area of the associated rectangle. Through decomposition, the area of a trapezoid can be specified in terms of associated rectangles, a challenge that Reflect 1.6 poses.

Reflect 1.6

Suppose that we decided to base all the area formulas for quadrilaterals on rectangles rather than squares. What would the formula be for a trapezoid, say, or for a parallelogram? For other quadrilaterals?

Similarly, relationships between the volumes of corresponding three-dimensional figures can be specified precisely and simply. It is quite feasible to stay close to the geometry and come to a far deeper understanding of area and volume—as *geometric* notions, rather than simply as a number of some units, calculated according to a formula.

The Importance of Imagery: Big Idea 2

Big Idea 2. *Geometric thinking involves developing, attending to, and learning how to work with imagery.*



Geometry, as a human endeavor, involves working on and with images. Human beings have the capacity—developable and educable in a variety of ways—to conjure images and to manipulate them. You are asked to do this in Reflect 1.7.

Reflect 1.7

Imagine a square. Now make a copy of it, turn it in your mind by 45 degrees, and place one of its vertices at the center of the first square. What is the shape of the intersection of the two squares?

If someone asks you to think of a square, you will almost certainly imagine one in your mind. How big is it? How is it oriented? Can you turn it around? Which center of rotation did you use? In which direction did you go? These are all decisions that you made for your own imagery of the situation—imagery on which you will base anything else that you are subsequently asked to do. Students in mathematics classes are sometimes challenged, either occasionally or intensively, to work on their mental arithmetic. How and when are they comparably challenged to work on their mental geometry?

In Reflect 1.7, you might have had some trouble imagining the overlap; you probably had to shift your attention to the relationship between the half-diagonal of the first square and the side of the second one. Or perhaps you imagined a simpler situation in which the second square was placed at a 90-degree angle, and then you mentally rotated it around to a 45-degree angle. These are both techniques of mental imagery that might not come naturally but can be worked on and developed, much as strategies for mental arithmetic can.

Where does geometric imagery come from? Natural objects in the environment are endowed with shape: the circularity of the cross sections of trees, the hexagonality of honeybee cells, the symmetry of fruit. Objects in nature have been shaped by a variety of physical forces; in the main, human-made objects have been designed and are shaped by the means of their manufacture. Shaped objects are therefore commonplace in the everyday world, a fact that goes a long way toward accounting for the dominant elementary school attention to shape rather than to the processes and purposes of shaping. But geometry also concerns itself with properties of shape, with relationships among the components (such as congruence or proportionality, parallelism or perpendicularity or

the situation of being somewhere in between)—with requirements that *force* certain things to be the way they are or that *prevent* them from being otherwise. There is a *compulsion* at the heart of geometry, a *must be* or a *cannot be* about the entire realm of *possible* configurations—possible, that is, in the imagination. For geometry constrains even our imaginations.

It is often when we are asked to draw shapes that we become more conscious of our mental images of them, as well as more aware of relationships and components. Consider the task in Reflect 1.8, adapted from Watson and Mason (2005), which draws attention to some of the ways that relationships among the components of shapes can determine and constrain them.

Reflect 1.8

Draw, one after another—

- a quadrilateral;
- a quadrilateral with one pair of sides of equal length;
- a quadrilateral with one pair of sides of equal length and one pair of sides parallel;
- a quadrilateral with one pair of sides of equal length and one pair of sides parallel and one pair of opposite angles of equal measure.

Many begin this task with a square or a rectangle. It is certainly common to start with familiar, “comfortable” quadrilaterals, as if they are somehow “more quadrilateral” than others that bear the same name. A rectangle is perhaps what comes to mind when you think *quadrilateral*—or perhaps it is just easier to draw. As you move through the sequence of questions in Reflect 1.8, you are forced to make sure that certain relationships hold. But you are also likely to become aware of your options: will the sides of equal length be opposite or adjacent to each other (and does that cover all the possibilities)? Will the sides that are parallel also be the ones that are of equal length or not? Which choice is easier to picture in your mind? Which is easier to draw? Is there any reason for opting for one over the other?

This task is meant to show how narrow our space of examples can be when we think about geometric shapes and how important it is to be able to have access to a wider range of examples. It also shows the value of thinking not only about the properties that a shape might have, but also about the ones that it should not have. Reflect 1.9 encourages you to explore the range of drawings that can satisfy the requirement of each step in the sequence, to create shapes that you perhaps have never seen before (let alone drawn), and also to work on expanding your imagery involving quadrilaterals.

Reflect 1.9

Return to the drawings that you made for the task in Reflect 1.8. Go back and see whether at each stage in your sequence of drawings your example does or does not satisfy the next constraint. If it does, produce a new example that does not fit the subsequent constraint.

The images and objects that geometry concerns itself with are not physical objects in the world, even though they can, at times, be inspired by them. They are not shaped by physical forces (such as the tides or the wind), but instead can be acted on mentally, shaped by human intention and desire. To that extent, they are closer to manufactured or designed objects. But nevertheless there are always constraints: with physical objects, these include the resistances of the medium and the materials (e.g., gravity, friction). With mental objects of shape, the constraints lie primarily in the definitions and theorems of geometry.

The development of imagery becomes very important in the transition from elementary to middle school. The concrete shapes of the elementary school allow tangible interactions, but also carry with them many unnecessary features: the three-dimensionality of the block, its roughness and color, the particular shape of the triangle or hexagon. In the move toward a more theoretical understanding of *triangle*, these features must take on far less importance. Even the drawing of a triangle on a piece of paper contains unnecessary thickness, color, and texture. Developing geometric imagery involves learning to see *through* drawn or concrete objects to the underlying geometric one. This does not simply require shifting *from* the visual to the verbal (from pictures to written definitions). It also involves going back and forth, many times, between the visual and the verbal.

The content of geometry

Essential Understanding 2a. *Geometric images provide the content in relation to which properties can be noticed, definitions can be made, and invariances can be discerned.*

We identify properties of shapes, make definitions, and discern invariances in relation to the content of geometric images. To explore this idea, we propose to continue working with quadrilaterals, no matter how familiar or overworked quadrilaterals may seem. Our attention to imagery will shift the focus from an identification-driven one (“Which of these shapes is a rhombus?”), or even a definition-driven one (“What is a rhombus?”), to a relational and variational one. Questions surrounding identification and definition can get in the way of developing imagery. For example, young children can



identify various shapes (such as triangle, square, circle, trapezoid), but their work on identification generally focuses on a very canonical set of examples and images. Their vocabulary words (*square*, *triangle*, *circle*, etc.) are not necessarily used to communicate the range of examples that such named concepts can include geometrically. Even such apparently simple concepts in plane geometry have a far greater richness than is ever realized in many textbooks or classrooms. By working here on the notion of quadrilateral, our primary intention is to exemplify ways in which that richness can be brought forward for discussion through working with images.

In what follows, we present a sequence of tasks aimed at exploring two general questions surrounding relation and variation in a particular context: “How different can a quadrilateral be and still be a quadrilateral?” and, “What makes a certain quadrilateral ‘special’?” We want to examine how these questions relate to geometric imagery, beginning with Reflect 1.10.

Reflect 1.10

Draw a quadrilateral that has nothing special about it whatsoever. As you do this, pay attention to how you decide that some feature is “special” and therefore needs to be altered (and how you go about doing that) or what features you are deliberately avoiding in order not to create a “special” one. Is it always clear how to change your drawing to make it “not special” while ensuring that you have not introduced a different “specialness” in so doing?



Big Idea 4

Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.

Geometric shapes that are endowed with names have something special about them—but there are many geometric shapes that do not have names! This is not always obvious to those new to geometry. What makes something special? Why does the quadrilateral with side lengths of 1, 2, 3, and 4 units not deserve to be named? What would make it worth naming? In this book, we will be returning frequently to what it is that makes properties, relations, or ideas about objects or situations “geometric” in nature—and hence, worth naming. Naming is also part of Big Idea 4, which we explore later in this chapter. In Reflect 1.11, attention is on what it means for a quadrilateral to be special.

Reflect 1.11

Consider your (probably unwritten) list of ways in which a quadrilateral can be special. For each feature on your list, draw a quadrilateral that has *only* that special feature and *none* of the other features on your list.

Sally’s list has the following as the features that make a quadrilateral special: two angles are the same, two sides have the same length, two sides are parallel, and the quadrilateral has reflectional

symmetry. To make a “non-special” quadrilateral, Sally chose to draw some quadrilaterals that had only two angles the same (see fig. 1.23).

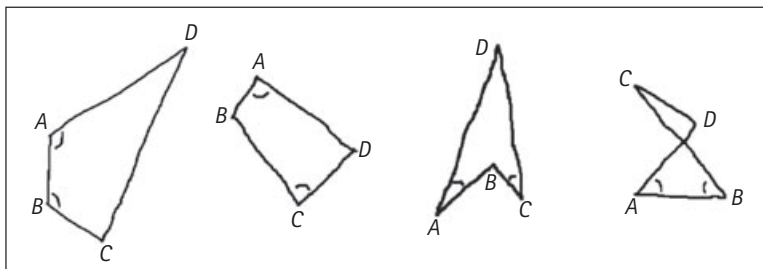


Fig. 1.23. Sally's drawings of quadrilaterals that have only two angles the same

Sally found that making the first quadrilateral in figure 1.23 was easy. The second was more difficult—after having drawn one quadrilateral with two adjacent angles the same, she wanted to try to draw a quadrilateral with two opposite angles the same. She had to be careful to not let AB and CD be equal. Neither case presented any difficulty in avoiding the reflectional symmetry feature. Sally made her third quadrilateral concave. She had the chevron in mind, but had to be careful to avoid any reflectional symmetry. For her final quadrilateral, she tried acute adjacent angles again and found herself drawing something surprising. Is this even a quadrilateral? Does the fact that it crosses itself make it special? Indeed, does it make it *too* special—that is, rule it out as a quadrilateral altogether? Who gets to decide, and for what reasons?

The answers to these questions depends, in part, on how we want to define *quadrilateral* and for what purposes. They also depend on whether it is worthwhile to distinguish “non-crossing” quadrilaterals (calling them *simple*, perhaps) from “crossing” ones (calling them *self-intersecting*, perhaps), or to go to the trouble of ruling them out explicitly in the definition of quadrilateral. What happens at the boundary of the two types, when a non-crossing quadrilateral becomes a crossing one? Making and seeing this set of quadrilaterals will provide Sally with some insight not only into how properties (special features) might relate to one another but also into how they might generate examples or possible examples that have never been seen before. If Sally comes across a definition of a rectangle as a quadrilateral with four congruent angles, she might conjure up any one of the images in figure 1.24.

The possibility of seeing a crossed quadrilateral as a rectangle might encourage Sally to wonder whether the crossed quadrilateral should indeed be considered special. A crossed quadrilateral, like that in figure 1.24, does not have some of the properties associated with rectangles, such as that all angles must be right angles or that

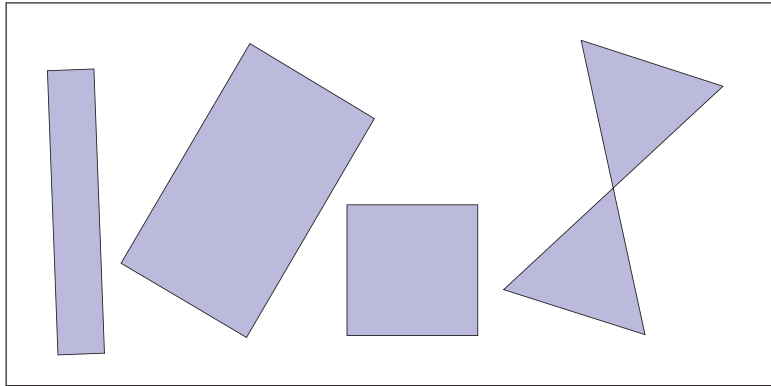


Fig. 1.24. "Quadrilaterals" with four congruent angles

any two opposite sides are parallel, or even that they have exactly four angles. The definition that we choose for a quadrilateral will depend on the conjectures that we want to make about them and even the theorems that we want to be true.



Cutting the work in half

Essential Understanding 2b. *Symmetry provides a powerful way of working geometrically.*

In our previous discussion, we observed that the last item on Sally's list of features that make a quadrilateral "special" was reflectional symmetry. She thought that a non-special (or non-specific) quadrilateral should not have any reflectional symmetry. Can there be reflectional symmetry without some sides or angles being congruent or without any pairs of lines being parallel?

Sally attempted to draw such figures, first with a pencil and then in a DGE. As she found herself needing to check angle measurements and side lengths, she discovered that she was unable to create anything that fitted the constraint. Sally decided to look at the problem in a different way. She tried to construct a symmetric shape by beginning with two arbitrary adjacent sides, segments AC and CB in figure 1.25, and then reflecting them across the line AB (to *force* the resulting quadrilateral to be symmetric). No matter how un-special she made her starting configuration, the result of the reflection guaranteed that $AC = AD$ and $BC = BD$, and moreover, both sets of opposite angles had to be congruent to each other. She had produced a kite, and her chosen method of construction had forced it to be one.

Sally's investigation suggests that it is possible to classify shapes not only by the familiar special properties (congruent angles, equal lengths, parallelism, etc.) but also by their symmetries. The drawing that Sally began with two arbitrary sides, shown in figure

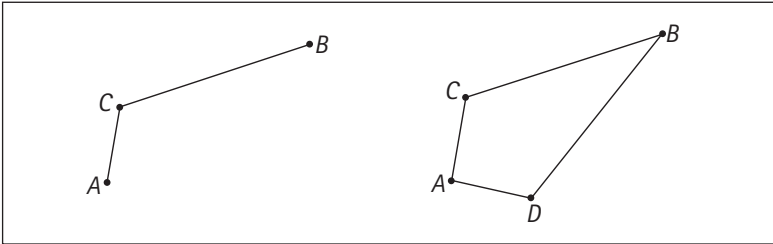


Fig. 1.25. Constructing a quadrilateral with reflectional symmetry

1.25, suggests quite quickly the possibility of making a shape with two types of reflectional symmetry, by simply making sure that $AC = CB$, thereby producing a rhombus. A first-level classification might thus involve differentiating shapes that have zero, one, or two lines of reflectional symmetry. Figure 1.26 contains Sally's table summarizing her results. Sally then wondered whether she had found them all and realized that she had forgotten the square and so would need an additional column in her table for four reflectional symmetries.

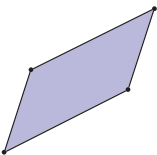
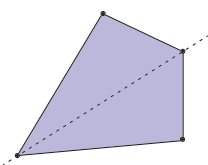
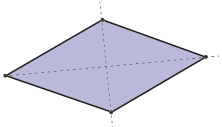
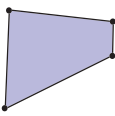
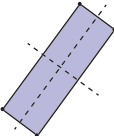
0 reflectional symmetries	1 reflectional symmetry	2 reflectional symmetries
Non-rhombus/ Non-rectangle Parallelogram 	Kite 	Non-square Rhombus 
Trapezoid 		Non-square Rectangle 

Fig. 1.26. Sally's provisional classification of quadrilaterals by reflectional symmetry

Sally's act of drawing the quadrilateral in figure 1.25 provides an immediate route from the class of one symmetry to the class of two symmetries, by starting with $AC = CB$. It also suggests how to create members of the class with no symmetries: in fig. 1.25, instead of insisting that $AD = AC$, one simply draws AD longer than AC (as Sally did), as if, say, to make a parallelogram. Reflect 1.12 invites you to consider the possibility of a quadrilateral with three lines of symmetry.

Reflect 1.12

The square has four lines of reflectional symmetry. Is there a quadrilateral that has exactly three? Are there other shapes that could be added to each of the columns above?

The two lines of symmetry for the rhombus pass through opposite vertices. For the rectangle, the two lines of symmetry are the perpendicular bisectors of the sides. Would it be possible to have a line of symmetry passing through opposite vertices, as for the rhombus, and just one line of symmetry that is a perpendicular bisector of a side, as for the rectangle?

Suppose that we begin with a rhombus (see figure 1.27). Our only option is to change the angle at a vertex (and once we change one angle, the adjacent one is determined). Label the midpoint of CD as G . The perpendicular bisector of CD is clearly not a line of symmetry. Let's change the angle at C to make it bigger. Now the perpendicular bisector of CD crosses AB but is still not a line of symmetry. In fact, it only becomes a line of symmetry (passing through the midpoint of AB) exactly when the angle at C is a right angle—that is, when we have a square. The only way to get at least three lines of symmetry is to have four! By starting with a rectangle, we could make a related experiment to show that the only way to obtain a line of symmetry passing through opposite vertices is by having all four sides exactly the same length.

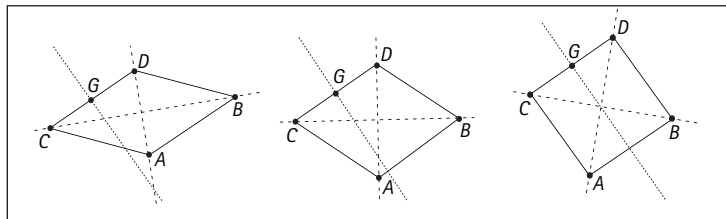


Fig. 1.27. Attempting to create a quadrilateral with three lines of symmetry

This exploration might suggest that one cannot get one line of symmetry without getting two. But the example of the kite clearly shows this to be false for lines of symmetry going through opposite vertices. Can there be a single line of symmetry passing through opposite midpoints? One example of such a shape is the isosceles trapezoid, whose line of symmetry passes through the midpoints of the two parallel sides. This shape would fit in the middle column of the chart in figure 1.26. Thinking dynamically, imagine the isosceles trapezoid getting taller and taller and narrower and narrower at the top until it eventually arrives at its limiting case. You have produced an isosceles triangle. If you started with an isosceles trapezoid with

base angles of 60 degrees, you would end up with a shape that does have three lines of symmetry: the equilateral triangle!

Are there other possibilities for one reflectional symmetry besides the isosceles trapezoid and the kite? We approach the question through construction in figure 1.28. If the shape is to have a line of symmetry through two opposite midpoints, we might start with two sides j and k . But we quickly realize that the segment joining their midpoints will not work as a line of symmetry; it must be perpendicular to both sides, as for l and m in the figure. But given l and m , which must be parallel to each other, the only way to complete the quadrilateral is to draw in the remaining sides o and n , thus producing an isosceles trapezoid. By scanning the collection of known shapes, we found that the isosceles trapezoid fit the middle column (although the *general*, non-isosceles trapezoid must remain in the column with 0 lines of reflectional symmetry). By constructing a shape that fits the middle column, we have found that the isosceles trapezoid is the *only* possibility other than the kite.

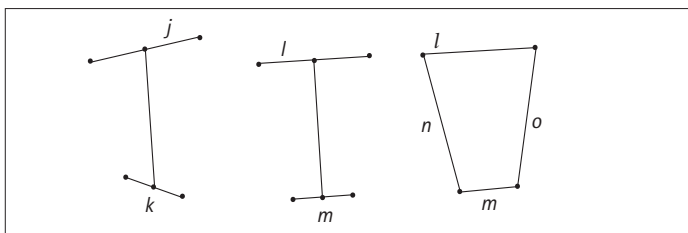


Fig. 1.28. The isosceles trapezoid

Like drawing, geometric construction enables one to work with the constraints and variations of configurations. Drawing focuses attention on one's hands and the physical reality of constraints that confirm or suggest properties and relations. Drawings in dynamic geometry environments complement handmade drawings by enabling actions that one's hands might never do (putting one vertex on top of another or on an already-drawn side; crossing one vertex across the figure; finding what are called "points of rupture" within a continuous drag), as well as producing a much cleaner and more accurate finished product. Constructions draw attention to what we might refer to as the "skeletal" elements of figures that determine their outer shapes. Reflect 1.13 encourages you to work with drawings to consider *rotational* symmetry for quadrilaterals in the way that we have previously considered reflectional symmetry.

Reflect 1.13

Create a table similar to the one in figure 1.26 for rotational symmetry for quadrilaterals. How many different types are there? How do the tables compare?

A quadrilateral has rotational symmetry if it overlaps itself exactly after being rotated around its center. Because all polygons return to their original configurations after a full (360-degree) rotation, all quadrilaterals (no matter how un-special) have at least one rotational symmetry. In thinking about how many types of rotational symmetry a quadrilateral has, it can be useful to attend to its angles. For example, consider the parallelogram. Rotating it by a half-turn (180 degrees) around its center (where the diagonals meet) will produce a perfect overlap. Because this is true for the parallelogram, we know it is also true for the rhombus. (We leave it to you to consider why.)

However, you may have thought that the rhombus would have more than just two rotational symmetries, given that its four sides are all congruent. Yet, consider the rhombus in column 2 of the table in figure 1.29, and imagine rotating it around by, say, a quarter-turn (90 degrees): it will not overlap itself, since a gray angle will correspond to a purple angle, and these two angles are not congruent. Because its opposite angles are congruent, a half-turn works and so does a full turn. The square has four rotational symmetries. You can rotate it by 90, 180, 270, and 360 degrees. Figure 1.29 captures the results of the exploration.

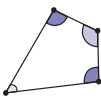
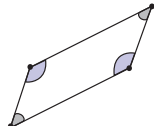
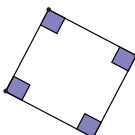
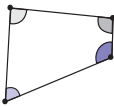
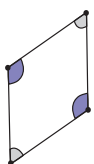
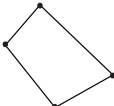
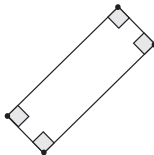
1 rotational symmetry	2 rotational symmetries	4 rotational symmetries
Kite 	Non-rhombus/ Non-rectangle Parallelogram 	Square 
Trapezoid 	Non-square Rhombus 	
Arbitrary Quadrilateral 	Non-square Rectangle 	

Fig. 1.29. Classifying shapes by rotational symmetry

Just as the square has the greatest number of reflectional symmetries, so it also has the greatest number of rotational symmetries. However, this time, the non-square rhombus and the non-rectangular parallelogram are in the same column, while the kite and the trapezoid are together in a different column. Combining information about rotational and reflectional symmetries can be a way of specifying particular shapes. For example, the non-rectangular parallelogram is the only quadrilateral with two rotational symmetries and zero reflectional symmetries.

Gaining geometric awareness

Essential Understanding 2c. *Geometric awareness develops through practice in visualizing, diagramming, and constructing.*

In addition to working with images through drawings and constructions, the mental mathematics approach—which usually refers to mental arithmetic—can be very powerful in developing and extending one’s mental imagery. Reflect 1.14 provides another opportunity for mental geometry.

Reflect 1.14

Imagine a rectangle. Picture its diagonals, and focus on them. How could you move the diagonals to turn your rectangle into a square? Now imagine a kite, and focus on its diagonals. How might you move these diagonals (a move that also, of course, changes the sides of the quadrilateral) so that the kite becomes first a square and then a more general rhombus?

For some people, the act of focusing on and moving parts of their body supports their visual imagery. In working through the task in Reflect 1.14, which asks you to imagine shapes with two diagonals, you can use your forearms to model the diagonals directly. You can make them move continuously, perhaps by translating one forearm across the other or rotating both about the point where they cross. If the task in Reflect 1.14 seems challenging, first use your arms with your eyes open, then repeat the motion with your arms with your eyes closed, and then finally keep your arms by your sides and simply visualize the movements that you made. This is an example of how everyone’s capacity to create and manipulate imagery can be strengthened and practiced, as indicated in Essential Understanding 2c. Reflect 1.15 provides practice with a different element of imagery—namely, the ability to make lines appear and disappear.



Reflect 1.15

Imagine a general convex quadrilateral with its diagonals drawn. Let the sides of the quadrilateral fade until they vanish, leaving only the diagonals behind, like the grin of the Cheshire cat in *Alice in Wonderland*. What do you see? Move the diagonals in some manner. Now make the sides of the (new?) quadrilateral appear. What new quadrilateral have you made by moving the diagonals?

Working in a DGE that has hide/show capacities can help develop imagery and offer unexpected configurations. Users can also see transitional configurations—for instance, where the two diagonals touch one another before becoming separated.

The task in Reflect 1.15 is fruitful because it turns upside down the idea of diagonals being derived *subsequently to* or *created from* the vertices of a figure. Our mental imagery in this situation has not been hijacked by the persuasive canonical examples that customarily come up when we think of squares or rectangles. The ends of the diagonals tell us where the vertices of the quadrilaterals *must* be, and once we have the vertices pinned down, it seems that the quadrilateral is fully determined, as long as the diagonals themselves cannot be sides of the quadrilateral. So we might want to say that not only are diagonals determined by the quadrilateral but that *quadrilaterals are (almost always) determined by their diagonals*.

What would be some consequences of *defining* a quadrilateral in terms of its diagonals (as one might define a circle in terms of its diameter rotated about the midpoint)? Definitions are neither fixed nor forced on us, but once they are selected or adopted, they have significant, forceful consequences. Reflect 1.16 explores how one might define a quadrilateral by means of its diagonals.

Reflect 1.16

Imagine two line segments. (How long are they? Do they intersect? Are they parallel?) Create a quadrilateral, using these two line segments as your diagonals. (Did you have to move them? What was the first quadrilateral you saw?) What properties does your quadrilateral have to have as a result both of the way in which you made your diagonals and the particular properties of the two line segments that you chose?

Starting from diagonals, we can imagine some new classes of quadrilaterals: the two line segments can be equal in length or not; they can cross in the middle of one diagonal, both diagonals, or neither; they can cross at right angles or not. These possibilities generate twelve related situations to explore. However, there are more situations than this; for example, the line segments can cross, not cross, or be parallel. If both segments bisect each other and are equal in length and cross at right angles, we claim that we must have a square (do you agree?). A parallelogram's diagonals must bisect each other, but they are not necessarily the same length (see fig. 1.30a).

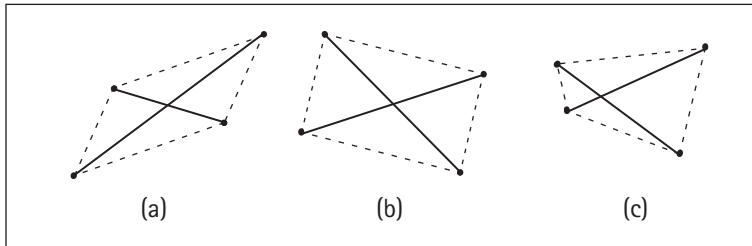


Fig. 1.30. Quadrilaterals determined by their diagonals

The diagonals of the rectangle, as shown in figure 1.30b, are of the same length and bisect each other. But what if we keep the diagonals the same length, without insisting on bisection, as shown in figure 1.30c? Does such a shape with these particular properties deserve its own name? What might other members of the class look like? Do you see any possible results that might be true about the entire class? Now that we have a hold on a property, we can try to imagine how we might change the shape, either in imagination or by constructing the diagonals in a dynamic geometry environment and dragging them around (see fig. 1.31).

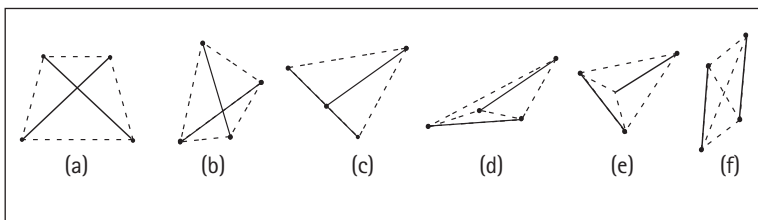


Fig. 1.31. Quadrilaterals constructed by dragging same-length diagonals in a DGE

The isosceles trapezoid comes out as a special case (see fig. 1.31a), one in which the diagonals are not only the same length but also intersect at the same relative point. In figure 1.31b, one diagonal bisects the other but does not produce a known (i.e., already named and characterized) quadrilateral. Figure 1.31c is a boundary example in which one diagonal is pulled along the first to its very end point. On what basis might this resulting shape be called a *quadrilateral*? By working explicitly with imagery, and in particular, with dynamic imagery, new possibilities can arise that encourage the forming of more precise descriptions and definitions for familiar shapes. This is in part the story told by Imre Lakatos (1976) in his book *Proofs and Refutations* with respect to the definition of a polyhedron and the claim that for some (which? all?) polyhedra, $V - E + F = 2$, where V is the number of vertices, E the number of edges, and F the number of faces (we describe his work in more detail in the discussion of Big Idea 4).

Big Idea 4



Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.

Extending the one diagonal beyond the tip of the other, as in figure 1.31d, produces concave quadrilaterals (or challenges your sense, and even definition, of *quadrilateral*, as the side-crossing possibility may have done earlier). The production of concave quadrilaterals through the variation of diagonals provides a very different characterization of them from the one typically offered—namely, that concave quadrilaterals have one angle greater than 180 degrees. Which is better? Why? By using a property of angles to specify concave quadrilaterals, we are always having to say something like, “One inside angle is greater than 180 degrees,” thereby having to specifically designate one of the two angles that any vertex can have as the “inside” one. This problem does not occur when we use diagonals as our differentiating property.

When one diagonal is placed at right angles to the other but without intersecting it, as in figure 1.31e, a more familiar shape is produced (here, a chevron, or perhaps this is also a kite?). Finally, figure 1.31f has diagonals that are parallel, which produces a crossed quadrilateral.

Focusing on the diagonals enables us to see more clearly the reflectional symmetry of the quadrilateral. Indeed, as figure 1.31 indicates, when we are working with images, we can attend to different parts or structures of the image, and this is an important part of working geometrically. Developing a geometric discourse involves attending not only to the number of sides or angles but also to the relationships between or among them. For example, when attending to relationships among the sides, we ask questions such as, “Are they congruent to one another?” or, “Are they parallel or perpendicular to one another?” Working in a dynamic geometry environment can make it easier to attend to relationships among parts or structures because they remain invariant under dragging. In fact, Battista (2009) has developed the *transformation-saliency hypothesis* to explain why people are so adept at noticing what stays invariant as they drag constructed shapes in DGEs.

Tools of the Trade: Big Idea 3

Big Idea 3. *A geometric object is a mental object that, when constructed, carries with it traces of the tool or tools by which it was constructed.*



Because the straight line and the circle are the most important objects of geometry—and far too common to overlook—we begin our discussion of Big Idea 3 with them. These objects do not exist anywhere in the world. We can construct them mentally or create models of them in sand, on paper, on screen. To do so, we need tools, which could be our own fingers, possibly chalk or pencils, maybe rulers or compasses, or a host of other instruments.

Drawing a straight line might seem pretty straightforward (no pun intended). Use a writing implement to sweep from one end of a ruler to the other to obtain a clean, seemingly straight line segment. But how do you know that the ruler is straight? If you were suspicious, how could you check?

Henderson and Taimina (2005) suggest a crease in folded paper as an excellent straight object. A ruler that is not straight would fail to lie evenly along the crease. Even tools that we take for granted can hide geometric complexities. Consider the use of tools in drawing a circle in the different ways suggested in Reflect 1.17.

Reflect 1.17

Draw a circle by hand on a piece of paper. Next, draw a circle that is smooth and round by using a compass. Then draw a circle by using a DGE. Pay close attention to two aspects of your work: what you are concentrating on with each of these tools and the order in which you do things with each.

In your freehand drawing, you probably began without much concern for the center of the circle (which was most likely hidden somewhere under your wrist), continuing on until you reached your starting point again (perhaps in one continuous sweep or in two or more arcs). The foremost aspect of a circle created this way is a closed, continuous path with constant curvature—and certainly not the radius. Your main challenge was to make your “circle” join up smoothly, and the length of its radius was probably predetermined mainly by the size of your hand.

In your circle drawn with a compass, you established the length of the radius first (as the distance between the two legs of the compass), and then you determined the center of the circle. You produced the circle itself by a rotation of the compass held by your hand—though it can still be challenging to produce an entire circle in a single sweep.

With the DGE, you chose the center of the circle first. A (very small) circle immediately appeared all in one go: you had no sense of a circle being generated over time as a locus of successive points. Instead, moving your cursor away from the center point generated a family of concentric circles, although only one was visible at any given time. You established the length of the radius only when you clicked the mouse a second time, having ceased dragging. There was no beginning or end point: your circle was seamless, singular, and complete. You could then stretch it or shrink it at will, almost as though it invited you to draw any circle.

Notice that none of these three ways of drawing circles features the radius itself as a geometric object. Reflect 1.18 suggests a different approach.

Reflect 1.18

Using the three tools involved in Reflect 1.17 (pencil, compass, and DGE), try the following tasks with each tool in turn: (1) place a point on your piece of paper (or computer screen), and then draw a circle that goes through that point; (2) find the center of your circle.

With the freehand circle, the first task is very straightforward since you can start your circle at the chosen point. The difficulty is with the second task. One way to locate the center involves realizing that the center point lies on every possible diameter of the circle. Thus, folding the circle exactly in half creates a diameter by means of the fold line. Repeating this process creates a second fold line that will intersect the first one at the center.

With the compass, the first task is not difficult but involves reversing the order of events described above; in this case, the pencil point is placed first, and both the center and the length of the radius can be freely chosen. The second task is completely unproblematic with a compass.

With the DGE, the center has to be chosen first but could be anywhere on the sketch (other than the given point), and the circle is drawn toward the given point until it lies on it. Again, the second task is completely unproblematic, as the construction makes use of the center point.

This rather lengthy introduction is meant to draw attention to the intimate link between the circle and the tools used to create it. Different tools—a pencil, a compass, a DGE—demand and permit different types of awareness of the circle. Geometric objects like the circle always have a history—a genesis by which they came into being—and sometimes that genesis gets imprinted in their definitions. Geometric objects are not everyday objects (like the lids of cans or the bases of cups) but are, in part, derived from them and then

extended or otherwise altered mentally. Consider the circle again. Although we are accustomed to seeing clean diagrams of circles on the printed page, we often forget how much the idea of a circle is intertwined with the tools used to create it.

Wellsprings of imagery

Essential Understanding 3a. *Tools provide new sources of imagery as well as specific ways of thinking about geometric objects and processes.*

Tools for geometry include paper, compass, carpenter square, linkages, blocks, ruler, grid paper, Miras, flipbooks, diagrams, screen images, human bodies. Tools are integral to geometric work, even though they are often forgotten. Reflect 1.19 asks you to connect tools to definitions.

Reflect 1.19

In Reflect 1.17, you explored three different ways of creating a circle, and each way involved a tool, which in turn highlighted certain aspects of the circle while de-emphasizing others. What is your idea of a circle? Give two different definitions of a circle. How do the tools implement these definitions?

When you are making a freehand drawing, your idea of the circle is probably related to the way in which your hand is turning the pencil to create the curved line that will end up at its starting position. Although you may not be thinking about this consciously, you are trying to maintain a constant curvature.

Another way of thinking about curvature emerges in work with the Logo programming language. In this environment, you can give an animated turtle instructions about moving forward or backward or right or left on the screen. To make the turtle trace a circle, you might direct it to move forward one unit and then turn right one unit, repeatedly. In fact, repeating the process 360 times traces out what appears to be a full circle. As in the case of the hand-drawn circle, the idea of a circle is less closely related to the center and the radius than to maintaining a constant amount of turn.

If you used a string with one end fixed and the other tied to a pencil that moved in such a way as to keep the string taut, you would have created a circle that strongly resembles Hero's definition of the circle, where the radius is the string itself:

A circle is the figure described when a straight line, always remaining in one plane, moves about one extremity as a fixed point until it returns to its first position. (Quoted in Bartolini Bussi and Boni [2003, p. 17].)



Hero's dynamic, mechanical, and procedural approach (saying what you have to do to create a circle, to bring one into being) contrasts with the approach of Euclid, for whom the compass had already become a mental tool. Euclid defined the circle in the following way:

15. A **circle** is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another.

16. And the point is called the **centre** of the circle. (Heath 1926, vol. 1, pp. 153–54)

The compass drawing of the circle strongly resembles the string method except that the actual radius is invisible, since no physical object relates one tip of the compass to the other. Euclid's definition of the circle bears the strongest resemblance to the DGE circle tool, in the sense that the center is mentioned explicitly, as is the actual curved line, which he called, and we continue to call, the *circumference* of the circle. In a DGE, the curved line appears immediately, and its radius can be changed, depending on where one chooses to click on the screen.

One final method that you may have chosen for drawing a circle involves the use of a carpenter's square. Here, you are running through all the possible right triangles with a fixed hypotenuse. You can easily find your circle's center, as the midpoint of the hypotenuse, but it is not immediately visible. However, you are using the property that all angles that a diameter of a circle subtends are right—a property that is not at all visible to someone using the other tools that we have mentioned.

Now that we've touched on triangles, let's take a look at how we construct a triangle of another common type. Consider Reflect 1.20, which draws attention to the isosceles triangle.

Reflect 1.20

Construct an isosceles triangle. How do you know that it is isosceles? What range of variation does it have?

The act of *constructing* involves identifying the geometric relations that will be specific to a particular shape. For the isosceles triangle, you want to make sure that two sides of the triangle are congruent. If you start with one side, segment AB , you can draw a circle centered at A and passing through B (see fig. 1.32). Any other radius of the circle will work as the second side for the isosceles triangle, since all radii of a circle have the same length. So, if you choose radius AC , then you can complete your isosceles triangle by drawing the segment BC . You know your construction is an isosceles triangle because segments AB and AC are radii of the same

circle. The range of variation might involve changing the radius of the circle or the angle at vertex A .

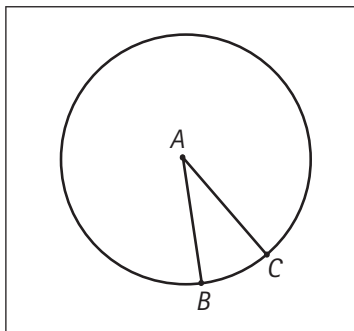


Fig. 1.32. Constructing an isosceles triangle

If you drew your circle by using a compass, you were using the compass not only to draw a circle, but also as a tool for creating a segment that would be the same length as another. The role of the compass tool would thus shift a little, in that it would become a tool for copying line segments. You might even have visualized the whole construction without actually using the compass. In this case, the compass would become more than a physical tool; it would become a mental one. This mental tool happens to be of great importance in geometric constructions (such as the equilateral triangle, but also the square and the rhombus). Being able to use it as a mental tool depends on having used it as a physical tool first. Further, you would have difficulty thinking of the construction of an equilateral triangle without imagining the compass tracing out its arc (and if you were to describe your construction to another person, your accompanying physical gestures would be likely to betray you). In this sense, the physical drawing of a circle with a compass gives rise to mental imagery that can be used in geometric thinking.

In a dynamic geometry environment, the circle tool replaces the compass as the means for both creating circles and copying lengths. But the compass (or the circle tool) is not involved just in producing circles. In fact, as one of the fundamental “implements” of Euclidean geometry, along with the straightedge, the compass is involved and invoked in constructing midpoints of segments, right angles, regular polygons, transformations, and so on. Among other things, what a geometric tool does is implement a definition. The whole development of Euclid’s *Elements* is based on these tools—if Euclid had chosen different ones, the geometry that we do (and teach) may well have looked quite different. The student who remarks with surprise (as many students do), “I never thought that you needed a circle to make a square!” experiences the lovely realization that the circle (through the compass) leaves its trace on the square.

Our discussion of the compass, in addition to this circle-making exercise, illustrates an essential understanding of geometry related to the role of tools in defining geometric *objects*. But tools also enable us to create new geometric objects. This process occurs repeatedly in geometry, particularly as physical tools become mental tools that can then be used to solve problems. In our ensuing discussion of Essential Understanding 3b, we look at how this process of moving from tool to object looks in the case of the geometric transformations studied at the middle school level.

The tool's the thing



Essential Understanding 3b. *Geometric thinking turns tools into objects, and in geometry the process of turning an action undertaken with a tool into an object happens over and over again.*

In addition to the compass–circle example, many other process–into–object shifts are relevant to middle school geometry. For example, the process of *rotating* an object around a point solidifies into *a rotation*, which is an object that has properties of its own. A rotation has a center of rotation and an angle of rotation; it is a transformation of the entire plane and, in particular, is an *isometry*. An isometry is a transformation that sends lines to lines and preserves many properties, including length, angle measure, and area. The transformation by isometry of one shape to correspond with another shape turns into the notion of figure congruence, which has properties of its own with regard to congruent sides and congruent angles. Various ways of cutting a doubled-napped cone by a plane result in curves specified by properties objectified as the *conic sections* (and the word *section* hides its Latin-verb origin: *seco* means “I cut”). One of the challenges of teaching geometry is keeping the dynamic, human actions—the verbs—alive in the relentless drift toward speaking only of mathematical things—the nouns—shedding connections along the way both to tools and to human interactions with them. Remembering that underlying the geometric objects are human tools is one way of contributing to this ongoing awareness.

Although the compass may seem to have a special status as a tool of geometry, many other tools have been and can be used as sources of mental imagery and objects. In the study of geometric transformations, for example, and of reflections in particular, mirrors (composed of a variety of materials, including the surface of water) and paper folding count as perhaps the most basic tools. Compare these tools with straightedge and compass as you respond to Reflect 1.21.

Reflect 1.21

How might paper folding, use of a Mira (see fig. 1.33), and construction with straightedge and compass differ in the way in which they express ideas about reflection?

A Mira is a small plastic tool used in creating a reflection of a figure and drawing its image by looking through the plastic.

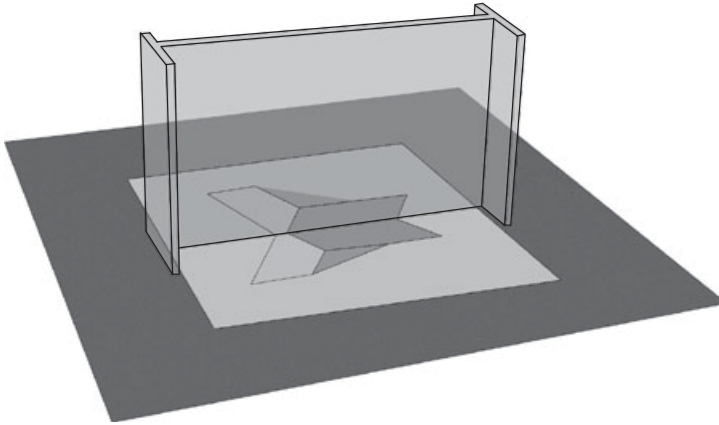


Fig. 1.33. Using a Mira to reflect shapes

To use paper to reflect a shape, the first thing to do is fold the paper to create the crease that will become the line of reflection (creasing is the action tool), while the next action is to rotate one part of the paper over that crease line until it coincides with the other. Thus, when carried out by paper folding, reflecting is about rotating in three dimensions! And the line of reflection must lie on the sheet of paper. Once the paper is flattened out again, however, the result can be seen as a pair of objects (the *pre-image* and the *after-image*, the latter often called simply the *image*), or the union of the pre-image and after-image is seen as one object with reflectional symmetry.

With the Mira, one also first chooses the line of reflection by setting the device down. But that line of reflection is much less permanent than a line of reflection in paper folding. Moreover, the Mira suggests not a *line* of reflection as much as a *plane* of reflection—making reflecting with a Mira also a three-dimensional process. But the image remains in the same plane as the pre-image, and there is a certain dynamism in the use of a Mira, since you can move it to the side, or up and down, or even on top of the pre-image. You are perhaps less likely to see the resulting configuration (both the pre-image and the image) as one object, since the Mira image is only a (typically reddish) shadow.

Construction tools require first drawing a line of reflection—that is, an actual line on the paper—and then proceeding point by point to construct the image of the shape to be reflected. This may be done in several ways. One is first to construct a perpendicular line to the line of reflection through, say, A in figure 1.34, and then to draw a circle centered at the intersection of the line of reflection and the line perpendicular to it passing through point A . The point A' at which the circle and the perpendicular bisector intersect for a second time will be the image of A . This construction uses the properties of reflection that A and A' are equidistant from the line of reflection and that the line of reflection is perpendicular to the segment AA' (expressed through the construction of the perpendicular bisector of the line through AA').

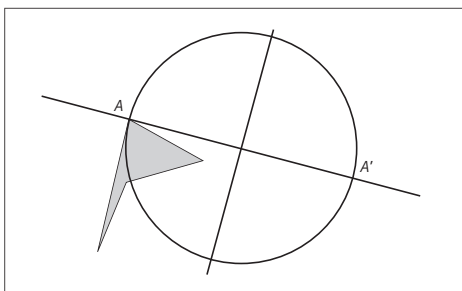


Fig. 1.34. Reflecting a single point of a shape by using construction tools

Unlike the use of the Mira and paper folding, the construction method of reflecting centrally involves using the defining properties of reflection. And in a dynamic geometry environment, that straightedge-and-compass process has been turned into a tool—namely, the reflect command in the transformation menu. Using this tool requires having a pre-image (an object such as a point, a line, a polygon, an ellipse, etc.) and also a line of reflection. Then, with the pre-image selected, the reflect command produces an image of the sketch in the same plane. In the DGE, the reflection does not have a three-dimensional component. As with the Mira, though, it is possible to move the line of reflection in any direction or to any distance from the object to be reflected. However, in the DGE, the image has the same status as the pre-image, in the sense that the objects of the image can be dragged, measured, traced, or treated in any way that the objects composing the pre-image can.

As with the paper construction, when looking at the DGE sketch, it is not evident which is the pre-image and which is the image, a situation that may draw attention to the fact that reflecting the image will produce the pre-image—there is a certain reciprocity. It also draws attention to the fact that the composition of a reflection with itself is the identity transformation (which leaves every point unchanged): reflecting a pre-image produces an image and

then reflecting that image across the same line of reflection produces the initial pre-image. In general, the composition of any two isometries—applying one isometry and then using the resulting image as a pre-image for another isometry—is again an isometry. You can convince yourself of this in the case of composing translations.

Sometimes the composition of two isometries of the same type can lead to a different type of isometry. For example, if you reflect a pre-image A to an image A' by using a particular line of reflection, and then you reflect the image A' by using a new line of reflection that is parallel to the first one, the combined transformation will be a *translation* of the first pre-image A to the final image A'' , as shown in figure 1.35. You might have been able to guess that the combined transformation would not be a reflection by thinking about the fact that the reflection transformation reverses the orientation of an object. So if you do this twice, you should be in the same orientation as you were at the beginning. It would thus be impossible for the composition of two reflections to be another reflection.

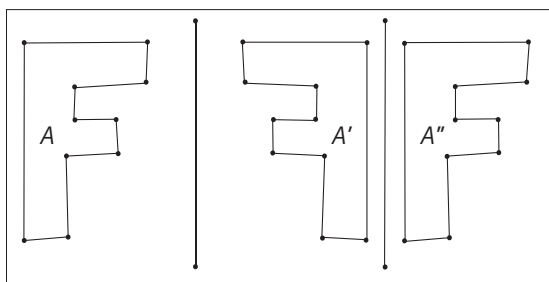


Fig. 1.35. The composition of two reflections can be a translation.

A slightly different approach, using a dynamic geometry tool, is to reflect just one point and then trace the paths of the point and its image, thereby simultaneously producing a pre-image and image in the plane. This use of the tool draws attention to behavior of the image point as the pre-image point moves (see fig. 1.36).

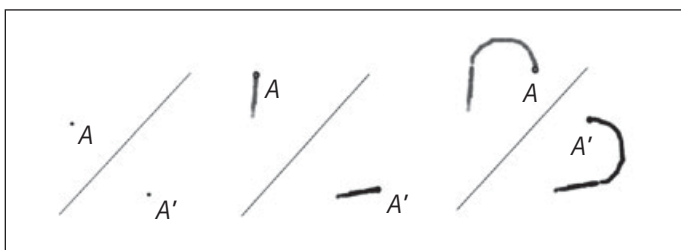


Fig. 1.36. Reflecting a point and then tracing both the paths of the point and its image at the same time

This approach might also lead to new questions, such as the following: “What happens when the point touches the line of reflection?” “Where are the locations where the two points, pre-image

→ Essential Understanding 3a

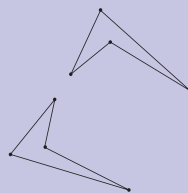
Tools provide new sources of imagery as well as specific ways of thinking about geometric objects and processes

and image, will coincide?” Reflect 1.22 poses questions about relationships among the image, pre-image, and line of reflection.

Once again, the tool mediates the action of reflecting, as described in relation to Essential Understanding 3a. What is of particular importance to the discussion of Essential Understanding 3b, however, is that the various processes of reflecting shapes eventually become objectified, through repeated reflecting, into the concept of a reflection. So, because reflecting R produced R' , we can speak of R' as a reflection (the product of the reflecting process) and, further, we can speak of the union of R and R' together as having *reflectional symmetry*. Although reflecting is a process applied to one shape to produce another, reflectional symmetry is a property of a single shape. Reflect 1.22 asks you to find a line of reflectional symmetry.

Reflect 1.22

With just the pre-image and image under a reflection showing, as illustrated at the right, how might you find the line of reflection? How would this method change according to the tool that you used to create reflections?

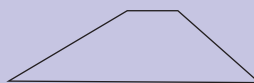


It is common in school geometry to investigate the reflectional symmetry of a given object, like a butterfly or a heart. To do so is to speak of a property of the object rather than about a process of transformation. When you are talking about reflectional symmetry and reflection, your interest is actually in the whole plane that contains the object (or objects) and not solely in the object itself. If paper folding is the tool that you use to find the line of reflection, you can think of the whole piece of paper as a representation of your plane. The status of the plane is less apparent if you use a Mira. In a dynamic geometry environment, the whole sketch area might represent the plane.

It is also common in school geometry to investigate the rotational symmetry of some objects. Consider the rotational symmetry of the quadrilaterals in Reflect 1.23, for example.

Reflect 1.23

Which quadrilaterals below have rotational symmetry? Do some have more rotational symmetries than others?



We focused earlier on the double move from *reflecting* (verb or action) to *reflection* (noun or thing) to *reflectional symmetry* (property), but we could also have had a comparable discussion about the double move from *rotating* to *rotation* to *rotational symmetry*. In identifying rotational symmetry, we might find it helpful to think of rotating the whole sheet of paper containing the shape around a fixed point (which one?). If we think in this way, will we “see” the shape coincide with itself?

Of course, rotating each shape around its center by 360 degrees will produce a perfect superimposition. The center of the rotation may be much easier to find for the square and the rectangle than for the parallelogram and the trapezoid, though! In fact, you can rotate any of these shapes by 360 degrees around *any* point, and your rotated image will overlap with the original! However, for the rectangle and the non-rectangular parallelogram, for example, the center of the rotation must be the center of the figure (the point where the diagonals intersect) for a rotational symmetry of 180 degrees. However, the trapezoid has no such point; it has only the trivial rotational symmetry of 360 degrees.

The idea that two-dimensional symmetries involve the whole plane is easier to appreciate by considering symmetries in a different context. For example, consider the border patterns shown in figure 1.37. Here, the domain is an infinitely long strip of paper. If you want to find a symmetry, you need to transform the whole strip. Reflect 1.24 asks you to identify and compare symmetries of border patterns.

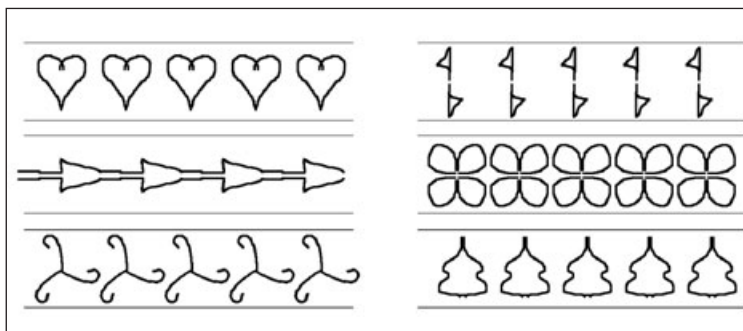


Fig. 1.37. Border patterns

Reflect 1.24

What symmetries does each of the six border patterns in figure 1.37 have? Are the symmetries of any one border pattern “the same” as the symmetries of another pattern? Can you think of border patterns that would have symmetries other than the symmetries of the patterns shown?

All of the above patterns have translational symmetry. For some people, this is part of the *definition* of a border (or strip) pattern. The first pattern in the left column in figure 1.37 also has reflectional symmetry, because the whole (infinitely long) border pattern could be reflected across any of the vertical lines that are the mirror lines for the individual hearts and the image would coincide with the original. This border pattern is “the same” as the third one in the right-hand column, in the sense that both patterns have only reflectional and not rotational symmetry and both patterns have vertical lines of symmetry.

The second border pattern in the first column has a horizontal line of reflectional symmetry through the center line of the strip. However, there is no such line of reflection in the third border pattern on the left. Does this pattern have rotational symmetry?

At first, the possibility of rotational symmetry in the third pattern on the left appears promising because each little design has rotational symmetry in the plane. However, if you rotate the border around the center of one of these designs—say, by 120 degrees—the one design and its image would line up, but the border pattern would no longer lie on the horizontal. So the strip as a whole has no rotational symmetry. The only symmetry that this pattern has is translational symmetry, since you could slide the strip of paper by a length equivalent to the distance between the center of each design (or any multiple of this distance in either direction parallel to the center line) and then get a perfect overlap.

A border pattern in which each design is different from the others will have no symmetry at all. (Again, for some, this is enough for it to fail to be a border pattern.) The first border pattern in the right column does have rotational symmetry, since it would be possible to rotate it by 180 degrees. We leave the second border pattern in the right column for you to consider, along with a challenge: find two more border patterns showing different possibilities for symmetry from those represented in figure 1.37.

In our investigation of border patterns, we have tried to show that there are different ways of transforming the pattern without changing it. In high school geometry, each of these ways of transforming becomes a thing to be studied—an object. Reflecting becomes a reflection. Rotating becomes a rotation. Translating becomes a translation. In each case, the verb describing the action is turned into a noun. In geometry, one needs to be able to work with both the “process” and the “object” ways of thinking.

Our discussion so far has focused on the isometries, but dilation is another type of transformation that is studied in middle school geometry. Figure 1.38a shows two similar shapes, and figure 1.38b shows these two same shapes along with the center of dilation that relates one to the other. Just as the notion of congruence

The set of all possible isometries includes *glide reflections* in addition to rotations, reflections, and translations. See *Developing Essential Understanding of Geometry for Teaching Mathematics in Grades 9–12* (Sinclair, Pimm, and Skelin 2012) for details.

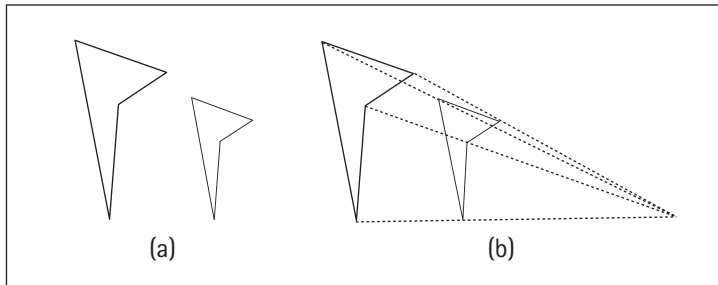


Fig. 1.38. Two similar shapes that are also related by a dilation transformation

connects strongly to isometries, so the notion of similarity connects strongly to dilations.

Although reflection and rotation have tools very clearly associated with them in school geometry, the dilation transformation does not. The pantograph, which was created in the early seventeenth century and is illustrated in figure 1.39a, is an excellent mechanical tool for copying as well as enlarging and shrinking drawings on the plane. Figure 1.39b shows a geometric model of a pantograph. The end of the pantograph at A would be fixed to the piece of paper. CD and BE are parallel, as are CB and DE . The angles at C and B can change. So if you trace B as you drag it to make a shape, F traces out a dilated shape. The longer CD is (with D still beyond C relative to A), the bigger the dilation will be. But if D is on AC , F will trace out a shape smaller than the initial one.

The Connected Mathematics Project (Lappan et al. 2006) suggests a simple tool for students to make and use in exploring dilations. The students tie two rubber bands together and hold one end fixed on a piece of paper at an anchor point. Then they slip a pencil into the loop at the other end (see fig. 1.40). The students then move the pencil in such a way that the knot follows the contour of

For more on devices that produce geometric drawings, including many to try to create a straight line, see Henderson and Taimina (2006) as well as <http://kmoddl.library.cornell.edu/tour/>.

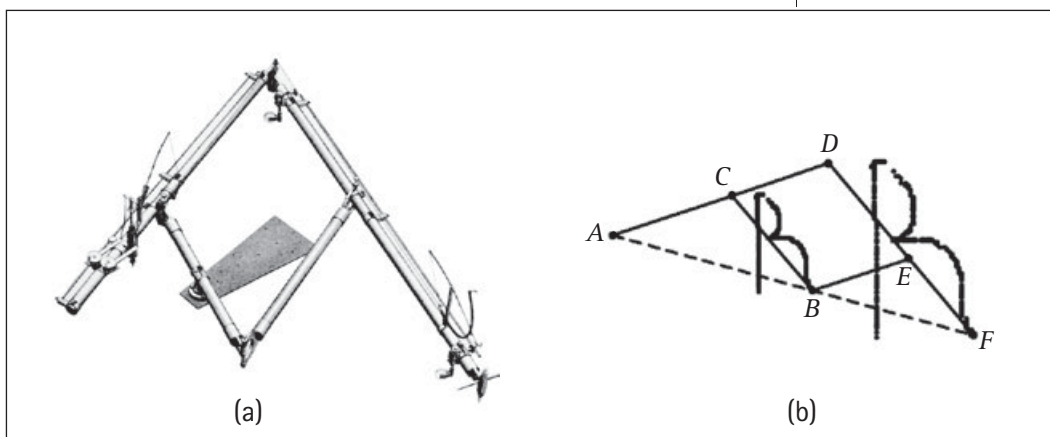


Fig. 1.39. A pantograph and a geometric model of the pantograph

a shape. The pencil thereby produces the dilated shape, with a scale factor of 2 and a center of dilation at the anchor point. This tool permits students to feel the effect of increasing the length of each line segment of the shape in the movement of the hand on the pencil. If they add more rubber bands, they can create dilations with different scale factors. Similarly, by using one rubber band and spacing knots on it, it is possible to create dilations with scale factors between 0 and 1.



Fig. 1.40. Rubber bands as a tool for exploring dilations

Everyday digital dynamic technologies also offer tools for dilation—think of the zoom features of cameras, word processors, and touch-screen devices, as well as the dilation and zoom tools offered by DGEs. These provide rich sources of imagery for dilation.

Working with a virtual tool that is similar to the rubber band device but constructed in a dynamic software environment (see fig. 1.41), the user can drag a point R around the contour of the shape while the endpoint S of the segment whose length is fixed draws the dilated shape. In figure 1.41, the scale factor is 2, and it can be changed directly, by clicking on the parameter. This move changes the length of segment RS relative to the length of the segment from the anchor point to R . Users can model dilations with the same center but different scale factors by dragging R around the contour of the shape after changing the scale factor, or they can construct the locus of S as R moves around the shape.

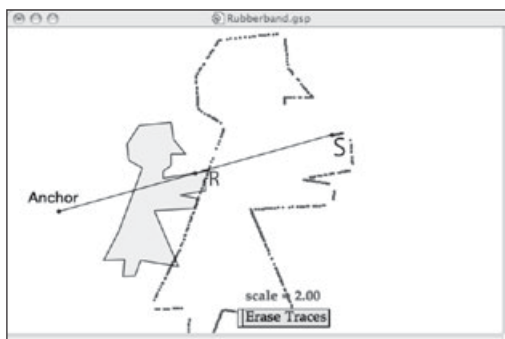


Fig. 1.41. A dilation in a DGE

In a DGE environment, unlike with the rubber band tool, it is a very simple matter not only to increase the scale factor (to 3, 4, or even larger factors) but also to change it to 1, 0, $\frac{1}{2}$, or even a negative number. Negative scale factors for dilations are always the most interesting ones, in part because they seem to defy physical rules.

In addition to facilitating changes in the scale factor, this dynamic model also enables the user to drag the anchor point. What will change when you move the anchor point? Many have a strong intuition that the size of the image shape will change. However, as shown in figure 1.42, changing the anchor point simply changes the position of the image, without affecting its basic size or shape. This makes sense, since the distance from the anchor point to S is still twice the distance from the anchor point to R .



Fig. 1.42. Changing the position of the anchor point

One way of deciding whether two shapes are similar is to consider whether there is a dilation point through which the lines connecting corresponding points would pass, as shown in figure 1.43.

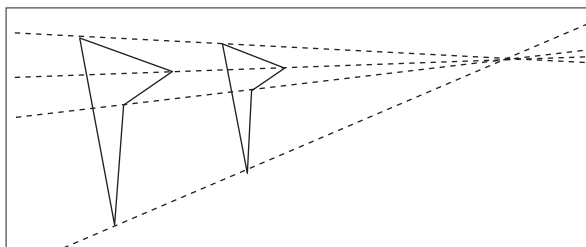


Fig. 1.43. Determining whether two shapes are similar by finding a center of dilation

Frequently, textbook questions on similarity present shapes that can be shown to be similar by using measurement but that are not as easily interpreted in terms of transformations. In fact, the two shapes in figure 1.44 are similar, but one could not use the process outlined above to find the center of rotation in order to locate a center of dilation.

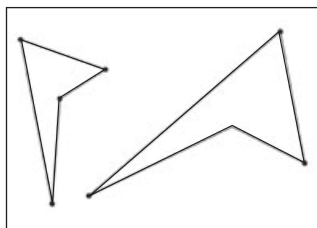


Fig. 1.44. Two similar shapes

Nonetheless, rotating the shape on the right around any one of its points until all corresponding sides are parallel, as shown in figure 1.45, would be sufficient to enable us to see one shape as a dilation of the other. Indeed, all questions about congruence and similarity can be resolved through transformations. The central advantage of working with transformations is to maintain a focus on the overall relationship between shapes rather than on the individual properties of each shape.

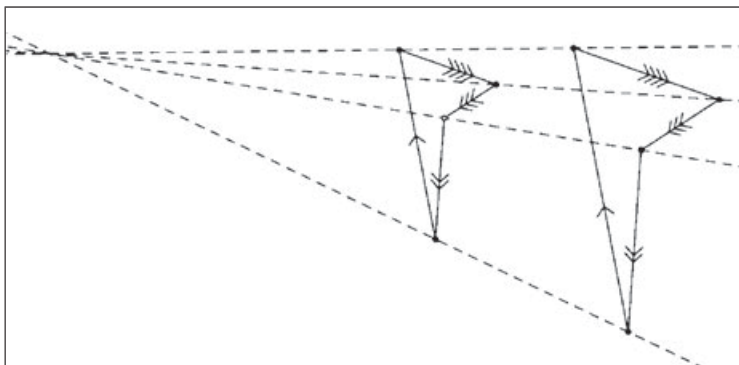


Fig. 1.45. Determining similarity

We have given detailed attention to dilations for several reasons. First, they illustrate the way in which the tool can affect the idea of dilating. The rubber band tool highlights the scale factor in relation to the number of rubber bands (or number of knots in a single rubber band). The dynamic geometry environment flattens the difference between whole number scale factors and real number ones by treating them all the same way. Further, we have seen how the tool of dilating produces a dilated shape, which is geometrically similar to its pre-image. Speaking about the geometric similarity between the pre-image and its dilated image involves forgetting about the center of dilation. Likewise, speaking about the congruence of a pre-image and its rotated or reflected image involves forgetting about the center of rotation or the line of reflection, respectively.

In this discussion of Big Idea 3, we have examined the way in which many geometric verbs turn into nouns and how we, in turn, understand processes as objects. The words *reflect* and *reflection* may seem almost identical, but they can have very different meanings in terms of the tools and the reasoning associated with them. In the discussion that follows of Big Idea 4, we will look more deeply at the role of names in geometry and, in particular, we will consider where names come from and what they might signal in relation to the properties, definitions, classifications, and theorems of geometry.

What a Geometric Investigation Involves: Big Idea 4

Big Idea 4. *Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.*

Geometry arises as the endpoint of geometric investigation. No one sequence of steps or tasks leads to that endpoint, although the six actions identified in Big Idea 4 are central activities along the way. Geometric investigation has a continuity that invokes each of these six intellectual actions at some point.

One difficulty with school geometry curricula is that the linking of the six actions to a specific investigation is often lost. Nouns replace the verbs embodying actions, which seem to have been carried out already and engraved in the textbook. In place of the verbs *classify*, *name*, *define*, *pose*, *conjecture*, and *justify*, we have the nouns *classifications*, *nomenclature*, *definitions*, *conjectures*, *theorems*, and *proofs*, all of which seem to predate the students' arrival in the classroom.

Notice that we did not list a parallel noun form for the verb *pose*, in part because the wondering (and even the wonder) that is so central to geometry surfaces so seldom in geometry texts, and this is one reason why these texts seem so set and certain. Also notice that we listed two parallel nouns—*theorem* and *proof*—for the verb *justify*.

Classifying: Geometry as natural history

Various classifications of geometric figures seem to flow through school geometry, including types of angles, types of triangles, and types of quadrilaterals. In this book, we have looked at different bases for classifications. For example, we classified quadrilaterals by diagonals, and separately, by symmetries. School geometry often requires knowing the classifications but not necessarily seeing classifying as a significant intellectual act. Students might not encounter classifying according to both a purpose and a specific basis or explore how different purposes and bases may give rise to a different classification.

Categories of figures identified in a classification tend to be named. Naming them identifies them as a single object with common properties or with a shared specification or definition. It may have come as a considerable surprise to find (on p. 37) that a type of quadrilateral specified by a property of its diagonals (that they be of the same length) does not have a familiar name.

The class of quadrilaterals with diagonals of the same length could, of course, be named—we baptize it here as “equidiagonal



Developing Essential Understanding of Geometry for Teaching Mathematics in Grades 9–12 (Sinclair, Pimm, and Skelin 2012) explores transformations in terms of invariance and investigates defining and proving in greater depth.

quadrilaterals.” (Notice in passing that whenever you and your students come across a new class of figures that does not have a conventional name, you are free to try your hand at naming it.)

Why does a name not already exist for this class? Turning things around, why is a shape like the *rectangle* already named? Like our “equidiagonal quadrilaterals,” was the rectangle too the result of an investigation of a geometric question and part of a classification? Did it too supply an answer to a geometric question?

One possible explanation for the namelessness of the quadrilaterals that we just dubbed “equidiagonal quadrilaterals” might be that this particular class of quadrilaterals does not generate any interesting results or theorems—what, if anything, can be said about *all of them at once*, across the full range of possibility? If nothing can be found to be invariant, then why bother naming them or even thinking of them as a single entity or type? Another reason for non-identification is that this class of quadrilaterals may not fit nicely into an inclusive hierarchy that relates it to other quadrilaterals. Of course, this last reason might not be defensible. *Cyclic quadrilaterals* (those quadrilaterals that can be circumscribed by a circle) are studied in high school, but they do not fit nicely into the typical school geometry hierarchy for quadrilaterals (Usiskin et al. 2008).

Our earlier diagonal investigation enabled us to attend to the range of shapes that a certain constraint might bring into being and the value of naming a particular class of shapes. The value is not in the actual vocabulary word itself but in the generativity of the constraint—namely, the fact that the “equidiagonal quadrilateral” represents a family of shapes sharing certain properties that apply equally to non-square rectangles and to squares.

Our exploration of diagonals is just a sample of what might be done with the quadrilateral: our use of it in this way was intended to be illustrative, not prescriptive. By beginning with a drawing, whether imagined, on paper, or on a computer screen, we can forge ahead, moving past the classified shapes of the school geometry canon to engage in the kind of investigation that lies at the heart of geometry. Classification is never complete, and the work of classifying shapes is much more centrally about evaluating the fruitfulness of a given classification than about finding a particular one and rigidly and exclusively sticking with it.

What's in a name?



Essential Understanding 4a. *Naming is not just about nomenclature: it draws attention to properties and objects of geometric interest.*

Naming may seem an unimportant part of geometry: after all, do you know anything worth knowing if you simply know the name of something? Surely, naming is merely conventional, arbitrary, the

result of someone somewhere in the past deciding what something was going to be called and then convincing others to follow suit. When do school students (or you, their teacher, even?) ever get to name something in geometry?

The physicist Richard Feynman (1981) was told by his father that he knew nothing about a bird if he knew only its name, leading Feynman to value “the difference between knowing the name of something and knowing something, which I learnt very early” (p. 5). We disagree with Feynman’s father. Frequently in mathematics, and in geometry in particular, naming both signals and reflects structure and classification and can indicate relationships among geometric figures or objects. Names draw attention to certain things: sometimes what they highlight is properties or attributes; sometimes it is their absence. “Languages ... are significant ... not in the sense that they determine the ways in which the members of the community perceive the world around them, but in the sense that they determine what the members of the community attend to” (Halliday 1978, p. 198). Sometimes specific words direct us toward visual or other similarities between geometric objects and real-world objects. Unfortunately, much of this resonance and word imagery is masked by the fact that the roots of a great many English mathematical words and expressions are in Latin or Greek. Whence does our English geometric vocabulary derive?

In the middle of the sixteenth century, Robert Record, in some of the first mathematics textbooks written in English, systematically set out to create a mathematical vocabulary based on Anglo-Saxon roots, rather than Latin or Greek ones. His efforts failed, in that his terms, by and large, were not taken up by others. But imagine how much easier it might be if tangents were called “touch lines,” equilateral triangles were “threelikes,” and acute angles were “sharp corners.” We are perhaps linguistically the poorer for this failure.

Moreover, the fact that a particular word is used in mathematics may also reveal the presence of a presumed theorem or result. We encountered an instance of this earlier when referring to “*the* base” and “*the* height” of a triangle and citing the formula for area as “one-half base times height.” Hiding behind our ability to talk about “*the* base” or “*the* height” as if these were unique attributes of a triangle is a theorem. This theorem says, reassuringly, that yes, although both “base” and “height” actually have three possibilities apiece, they always come in linked pairs—a particular base with a particular height. And surprisingly—perhaps astonishingly—it does not matter which pair we use: the end result will always be the same.

Finally, there is always some basis for inquiring into why something has the name that it does. Nevertheless, Big Idea 4 is different from the three big ideas that have gone before it. We



Distinctions between the two ways of defining mathematical objects, as described by George Molland (1976), are explored in greater depth in *Developing Essential Understanding of Geometry for Teaching Mathematics in Grades 9–12* (Sinclair, Pimm, and Skelin 2012).

certainly believe that it earns its space. But it is still worth asking, “Does knowledge of names in geometry count as geometric knowledge?”

Definition, naming, and classification

Essential Understanding 4b. *Definition can both generate and reflect structure: definitions are often dependent on a specific classification.*

There are two common means of defining a geometric figure: defining by saying what must be done to bring the figure into existence (*defining by genesis*) and defining by specifying the particular property that must be true for every instance of that figure (*defining by property*). Naming is often related to definition and can reflect one or the other of these means of defining.

Geometric discourse allows a type of naming that is very uncommon in everyday life, but it is one that enables the creation of hierarchies inside classifications. Since the requirements specified by one definition (e.g., of a square) may be an extension of the requirements appearing in another definition (e.g., of a rectangle), one and the same shape may now be called by two different names (e.g., both *square* and *rectangle*). Similarly, something that might look like a straight line may also be called a *circle*, in the sense that the line is the locus of points equidistant from a given fixed point that is infinitely far away. Consider, for example, starting with a circle as shown in figure 1.46 and moving its center up, farther and farther away.

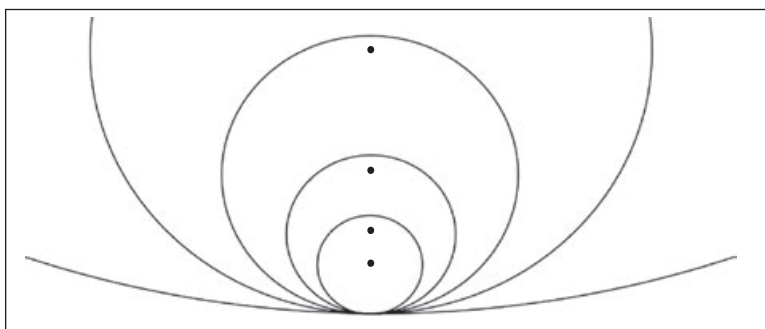


Fig. 1.46. The center of a circle moving further and further away

Trapezoid (in North America, at least) usually refers to a quadrilateral with one pair of opposite sides parallel. Authors of different textbooks may require *exactly one* pair to be parallel (with the result that a parallelogram is not a trapezoid, by definition), while others require *at least one* pair of opposite sides to be parallel (whereby a parallelogram *would* be a type of trapezoid). Defining *trapezoid* one way or the other has mathematical consequences,

including the classifications and inclusions or exclusions that would occur as a result. The basic idea of both definitions is an added property requirement on at least one pair of opposite sides—namely, that they be parallel. Explore the properties of the special trapezoid given in Reflect 1.25.

Reflect 1.25

Explore some geometric properties of the following very special type of trapezoid: three of the four sides are the same length, and the fourth side is twice as long as each one of the other three.



Isosceles trapezoidal tables with the short side of the tabletop half the length of the long side are increasingly common furniture in schools. They have various properties worthy of mathematical exploration (e.g., space-filling properties, properties of symmetry, properties of the diagonals). In thinking about this shape, you may immediately start to classify it as a special kind of trapezoid, and, indeed, a special kind of isosceles trapezoid. But there are reasons why we might want to assign it its own name. Unlike other trapezoids, this one is half a regular hexagon and composed of three identical equilateral triangles.

We can create new polygons that do not yet have names by seeing how they relate to existing polygons and finding whether they have any interesting properties. We can also give new names to familiar shapes that are based on different sets of properties. Reflect 1.26 provides an opportunity to rename a figure.

Reflect 1.26

Think about the diagonals of a geometric kite. In the simple toy of the same name, the two crosspieces seem to cross at right angles, and the longer one bisects the shorter one. Is that true for a geometric kite? How might you rename the kite to give it a name more in keeping with its geometric properties?

Reflect 1.26 asks you to focus on the properties of the kite in renaming it. If we now think about the names of types of triangles, for instance, we find they are named by the properties that they exhibit or lack. However, the names *scalene*, *isosceles*, and *equilateral*, coming as they do from Greek and Latin, mask this attention to properties.

Scalene and *isosceles* look like they might have some etymological connection with each other, but alas, they do not. The Greek word *skalenos* means “uneven,” and hence, “unequal”: the name *scalene* signals that there is no relation among the side lengths. A scalene triangle is so named for what it lacks—for what is missing. The Greek prefix *iso-* means “same” (as in *isometric*, *isotope*, *isobar*), and *skelos* means “leg,” while *equilateral* has Latin roots and means “equal-sided.” Why should *isosceles* refer to a triangle having exactly two sides the same length while *equilateral* means that all three have the same length?

The only reason that we can see comes from the use of the Greek word for *leg* in *isosceles* and the fact that humans have two legs. This use of *legs* to mean “two legs” continues in speaking of the “legs” of a compass (which typically have the same length) and the “legs” of a right triangle (which decidedly do not). The fancy-sounding set of terms for triangle types specified by property is really as follows: *scalene* for “uneven” or “lopsided” triangles, *isosceles* for “same-legged” triangles, and *equilateral* for “equal-sided” triangles.

One consequence of naming something is that it draws attention to things that do *not* suit that name, which in effect turns them into things to consider. In the case of triangles, naming *isosceles* and *equilateral* triangles draws attention to the undifferentiated mass of scalene triangles named solely in terms of what they are not.

In Reflect 1.26, you may have found that all geometric kites have diagonals that are perpendicular to each other, with one bisecting the other. In other words, the diagonals look like a lower-case *t*, which is suggestive of a new name for the kite—a “t-quad”!

We have already looked at one kind of classification of triangles (scalene, isosceles, equilateral). Of course, we can also classify angles into different types, such as *acute*, *right*, *obtuse*, *straight*, *reflex*, *full*. The word *angle* comes from the Latin *angulus*, a diminutive form meaning “little bend” (and relating to *ankle*). The sides of an angle are sometimes called “arms” and sometimes “legs.” *Acute* means “sharp” while *obtuse* means “blunt” (which were Record’s English choices). Notice that these ways of classifying angles can lead to a different classification of triangles: *acute triangle*, *obtuse triangle*, *right triangle*. Behind this naming sits some geometric knowledge (for example, that there can only be one right angle in a triangle and only one obtuse angle in an obtuse triangle). If we speak of an acute triangle, it is less clear whether it means having one acute angle (which, in fact, any triangle must have), two acute angles, or three acute angles. To speak of a *reflex triangle*, just as to speak of a unicorn, is to name an impossibility—at least in the plane (though reflex triangles do happily exist on the sphere, as do triangles with two right angles).

Like scalene triangles, irrational numbers are named for what they are not, as discussed in *Developing Essential Understanding of Rational Numbers for Teaching Mathematics in Grades 3–5* (Barnett-Clarke et al. 2010).

Moving on to quadrilaterals, we can begin with the term *quadrilateral* itself: “quadri-lateral” is a Latin term meaning “four-sider.” But what are the terms used to describe types of “four-siders,” and what are the properties that they signal? Further, why do we not call them “quadrangles,” given that we call triangles “tri-angles”? Why switch from number of angles to number of sides, and why does this work as a basis for naming geometric figures?

Square, rectangle, parallelogram, kite, trapezoid, and rhombus name different types of quadrilaterals. What structure or basis for naming can be seen in this variety of terms and their root meanings?

A *square* is the most restrictive type of quadrilateral. But, unlike for the most restrictive case of triangle—an equilateral triangle—requiring a quadrilateral to be equilateral is not sufficient if we want a shape that is also equiangular. A rhombus is the name for an equilateral quadrilateral, but not every rhombus is an equiangular quadrilateral (since an equiangular quadrilateral is a square, and not every rhombus is a square). In contrast, an equilateral triangle is equiangular as well as equilateral. We cannot have a triangle with one of the two properties and not the other. To determine that a quadrilateral is a square, we must verify that the quadrilateral is both equiangular *and* equilateral. Consider other properties of an equilateral triangle in Reflect 1.27.

Reflect 1.27

In any equilateral triangle, the common angle must be 60 degrees. Why is this so? In an equilateral and equiangular quadrilateral, why must the common angle be a right angle?

A rectangle’s name suggests that it has right angles (or perhaps just one?). Perhaps the term *rectangular quadrilateral* would be more accurate, as the figure’s four-sidedness has apparently dropped away from the name, and the property adjective (*rectangular*) has become a noun—the thing itself—a *rectangle*. There are several ways of thinking about why the angles of an equiangular quadrilateral must be right. One depends on knowing that the sum of the angles of any quadrilateral is 360 degrees, which means that each one would have to be 90 degrees.

Drawing on the angle-driven definition of rectangle, we might similarly propose angle-based names for other quadrilaterals, such as the parallelogram. A *parallelogram* takes its name from a property (parallelism) but without incorporating four-sidedness in the name. As a suffix, *-gram* means something that is written or drawn, as in *telegram*, *cardiogram*, *diagram*. For geometric figures, besides *parallelogram*, we have *tangram* and *pentagram* (a regular pentagon

with all its diagonals drawn in as well). The root meaning of *parallel* is “beside one another,” so a *parallelogram* is a “side-by-side-drawing.” In terms of angles, the parallelogram has (indeed, must have) two pairs of opposite, congruent angles. Notice that we have moved from the quadrilateral with four angles the same to the case with two pairs of angles the same. Are there quadrilaterals with exactly three angles the same?

Attention to different properties can give rise to other names for figures. Consider renaming a parallelogram, as suggested in Reflect 1.28.

Reflect 1.28

The diagonals of a parallelogram bisect each other, but do not necessarily have the same length. If they do, then the parallelogram is a rectangle. How might this property be used as the basis for renaming a parallelogram more informatively?

Why do pentagons not get the same detailed treatment as quadrilaterals in school geometry? Perhaps the four sides (or four angles) of a quadrilateral result in just enough diversity of “special” shapes, whereas the pentagon’s variety (two sides the same length and the rest different; three sides the same length and the two others different; three sides the same length as one another and the other two the same length as each other, and on and on—the pentagonal full house, so to speak) results in too much diversity.

So far, we have focused primarily on well-known two-dimensional shapes, but these shapes are made up of other, more basic geometric objects. For example, angles are often seen as components of a geometric figure rather than as geometric configurations in their own right. The Greek suffix *-gon* refers to angle. What if we called an angle a “gon” or an “enagon” (a “one-angle”)? Reflect 1.29 encourages you to pursue this naming structure.

Reflect 1.29

What could a “di-gon” (a “two-angle”) be? Note that this time, instead of asking you for a new name for a shape, we have used a naming structure to generate a possible name, and we are considering what geometric object might go with it.

The everyday mathematical term *diagonal* connects to this discussion. The Greek *diagonios*, from which *diagonal* originates, means “from angle to angle.” Although a common definition of diagonal speaks of a line segment connecting vertex to vertex, the Greek meaning refers to connecting one angle to another. How does thinking of the diagonal in this way change your mental picture of, say, the diagonals of a quadrilateral? Does it make it easier or

harder to accept a diagonal that lies outside the quadrilateral (as in the case of the chevron)? Perhaps the image of connecting two angles with a diagonal that lies outside the quadrilateral draws your attention to the fact that at any vertex of a quadrilateral, there are actually two angles.

Some of our discussion of naming may appear somewhat playful, even though we have tried to point to the ways in which naming can reflect structure. But related to our Big Idea 2, naming is strongly related to imagery. When you are asked to think of a quadrilateral, you may attend more to the sides of your mental shape than to its angles. If so, this changes the kinds of mental examples that you generate and the way in which you will go on to solve problems about, for example, quadrilaterals. Exploring the meaning of names, or generating new names, can be a way of working on imagery.

The decision about including or excluding particular properties or features is not simply a matter of choice—of tossing a definitional coin, as it were. The theorems that one wants to be true or to have a simple form also come into play. Consider the fact that a common theorem relates the volume of a pyramid (no matter which polygonal base) to that of the corresponding prism: the former is always one-third of the latter. One argument for including the cone as a type of pyramid and the cylinder as a type of prism is that the identical volume relationship also holds for cones and cylinders: the volume of a cone is also always one-third of that of the corresponding cylinder (see also the comments at the end of the discussion of Big Idea 1 earlier in this chapter).

Prisms and pyramids account for most of the three-dimensional shapes that are studied in school geometry. For all of these shapes, we have eighteenth-century Swiss mathematician Leonhard Euler's formula $V - E + F = 2$, where V is the number of vertices, E the number of edges, and F the number of faces of the solid. Note that in this formula we find a new word, *edge*, which is not part of the vocabulary of two-dimensional geometry (where *side* is used). *Edge* is needed to refer to the place where two faces meet. In fact, it was only after Euler identified this numerical relationship that the need for the new concept arose. Euclid, for instance, did not talk about the edges of polyhedra two millennia earlier.

What are some possibilities for three-dimensional shapes beyond prisms and pyramids? This is the question that Reflect 1.30 invites you to consider.

Big Idea 2



Geometric thinking involves developing, attending to, and learning how to work with imagery.

Reflect 1.30

What other three-dimensional shapes can you think of that are not prisms or pyramids? Which of these count as polyhedra?

We have already mentioned the cylinder and the cone, which not only fit uneasily into the categories of prisms and pyramids but also jeopardize the relationship $V - E + F = 2$ for all prisms and pyramids. In part, this is because it is not clear whether we can count curved surfaces as faces, nor is it clear whether, for example, the cylinder has any vertices. And what about a sphere? Or a stellated octahedron? Part of the mathematical story that Imre Lakatos (1976) tells in his book *Proofs and Refutations* is that the meaning of the geometric name *polyhedron* has changed over time. It started life as a rather naïve word referring to three-dimensional shapes with many different bases (*hedra*), but it could even be turned into a term meaning any three-dimensional shape for which the relation $V - E + F = 2$ holds (a curious definition by property).

For shapes such as the cylinder, for example, one has to be careful about what it means to be a face, a vertex, or an edge. If one thinks of the cylinder as being made of two circular faces (the top and bottom) and one rolled-up rectangular one, then the cylinder has three faces. The circles meet the rectangle in two edges, but perhaps the rectangle meets itself at an edge as well, so maybe there are also three edges. But there are only two vertices—the two places where edges meet. Thus, with this way of counting, $V - E + F = 2 - 3 + 3 = 2$ (see fig. 1.47). But *must* we count this way? To know how to count, one must first know what counts.

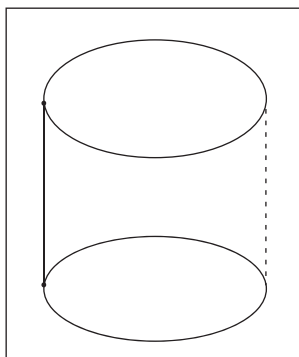


Fig. 1.47. A cylinder

This discussion of Essential Understanding 4b has pointed to the way in which names can arise from properties, including ways of generating shapes. Names can be interpreted in different ways and be sensitive to conjectures and theorems that are made about the shapes they describe. Because this is so, we can return to Feynman's claim mentioned at the beginning of our examination of Big Idea 4 and can say that knowing names—and knowing about where they come from and how they change—is quite relevant in geometry. Next, in discussing the final essential understanding for this book, Essential Understanding 4c, we shift our attention from the particular way in which names are used in geometry to the way

in which one's use of a name may change with different levels of geometric sophistication.

Where do conjectures come from?

Essential Understanding 4c. *Conjectures can emerge out of a problem-posing process that generates claims that need to be justified.*



In the previous sections of chapter 1, we have explored three different but related ways of thinking about quadrilaterals, each one of which gave rise to a specific classification scheme. You have known these names and have been familiar with the classifications since you yourself were in middle school. One of these classification schemes was based on diagonals. Others were based on reflectional and rotational symmetries, respectively. In our discussion of Essential Understanding 4c—our final essential understanding—we now draw on the work that you have already carried out to focus on conjectures and to illustrate how geometric investigation involves *classifying, naming, defining, posing, conjecturing, and justifying*.

When we made our classification of quadrilaterals according to their reflectional symmetries, we came up with four types of quadrilaterals. These types of quadrilaterals had no conventional names, a circumstance that leads to the question in Reflect 1.31.

Reflect 1.31

What might you name the four types of quadrilaterals classified according to their reflectional symmetries, and why?

Having come up with the four types of quadrilaterals—namely, those with 0, 1, 2, or 4 lines of reflectional symmetry, we posed the very reasonable question of whether there might in fact be a quadrilateral having 3 lines of reflectional symmetry. Having posed the question and conducted some geometric investigation by means of diagrammatic and visual thinking, we conjectured that no such quadrilaterals were possible. We then started to consider what a justification might look like by examining the properties that such a shape would have to have and explaining why that shape could not also be a quadrilateral. Notice that in the preceding example, the six central aspects of geometric investigation all came into play.

The previous example involves reflectional symmetry; Reflect 1.32 offers an opportunity to explore the same question with respect to rotational symmetry.

Reflect 1.32

Earlier, we classified quadrilaterals according to reflectional symmetries (see fig. 1.26) and rotational symmetries (see fig. 1.29). On the basis of our reflection classification, we considered whether any quadrilateral might have three lines of reflectional symmetry and determined that no such quadrilateral could exist. Now, on the basis of our rotation classification of quadrilaterals, let's pose a similar question, "Might there be a quadrilateral with three rotational symmetries?" Develop a conjecture, and start to justify why you think it holds.

Earlier in this section, we considered a new name for the kite based on its diagonals. We stated that the diagonals were always at right angles to each other, with one bisecting the other. But we didn't justify this claim. We now go back and state the claim as a conjecture, which we can try to justify on the basis of the more conventional definition of a kite—namely, that it has two pairs of adjacent, congruent sides. In the case of figure 1.48, diagonal AC creates two triangles, ABC and ADC . These triangles are congruent to each other, since all their pairs of corresponding sides have the same length. That means that their corresponding angles are congruent. Further, diagonal AC is the line of reflection that gives us triangle ABC and triangle ADC as reflection images of each other. This means that since D is a reflection of B , segment DB must be perpendicular to segment AC , and segment AC must bisect segment BD .

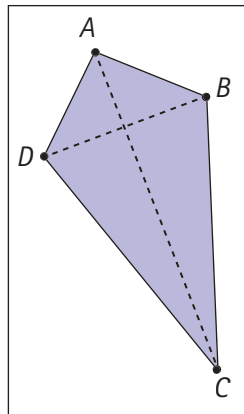


Fig. 1.48. Proving that the diagonals of the kite are perpendicular

The conjecture about the diagonals of a kite might instigate the development of a similar conjecture about the diagonals of a rhombus. You might also find a method of using reflection to justify the relationship between the diagonals of a kite fruitful in considering the rhombus.

Conclusion

We end chapter 1 by noting that the English word *theorem* comes from a Greek word meaning “vision.” As mathematician René Thom (1971) has observed, “According to a long-forgotten etymology, a theorem is above all an object of a vision” (p. 697). However, these are visions that are to be had in the mind; *vision* in this context does not refer to the ability to see. Working with geometric imagery involves not just sight, but encouraging certain sorts of images, not all of which have to be actually drawn or seen.

Our four big ideas for geometry have focused on the geometric underpinnings of measurement formulas, the importance of visual imagery, the way in which tools construct geometric objects in particular ways—as well as the inevitability of tools in geometry—and the unified nature of investigation in geometry, drawing on classifying, naming, defining, posing, conjecturing, and proving. In the next chapter, we turn our attention to the ways in which these big ideas interact both “horizontally” and “vertically” with the school mathematics curriculum.

Connections: Looking Back and Ahead in Learning

As the NCTM (2000) Content Standards indicate, geometry is an area of mathematical study both in its own right and because it is connected to ideas and problems across mathematics. Connections emerge through the use of imagery and diagrams, as well as specific ideas and results. In this chapter, we first look at how the big ideas from chapter 1 extend horizontally to other areas of mathematics in the middle school curriculum. Then we focus on looking vertically at how the big ideas grow out of and help support the geometry curriculum in both grades 3–5 and grades 9–12.

Looking Horizontally at Middle School Geometry

Proportionality is a major concept in grades 6–8. In this section, we connect ideas about proportionality that students encounter as they study ratio and slope to essential understandings of geometry discussed in chapter 1.

Ratio, proportion, and scaling or dilation

Many connections exist between geometry (and the particular essential understandings described in chapter 1) and ratio, even though these mathematical topics may seem very distinct. Lobato and Ellis (2010) identify one big idea in their Essential Understanding volume on ratios, proportions, and proportional reasoning: “When two quantities are related proportionally, the ratio of one quantity to the other is invariant as the numerical values of both quantities change by the same factor” (p. 11). This big idea points to some of the connections between ratio and geometry. Big Idea 2 of this geometry book draws attention to the central role of invariance in geometry. But much of geometry also involves the idea of ratio—of how one thing compares with another “in respect of size,” as Euclid puts it in Book V of his *Elements* (Heath 1926, vol. 2, p. 114).

Big Idea 2



Geometric thinking involves developing, attending to, and learning how to work with imagery.

According to Euclid,
"A ratio is a sort of
relation in respect
of size between two
magnitudes of the
same kind" (Heath
1925, vol. 2, p. 114).

See *Developing
Essential
Understanding of
Ratios, Proportions,
and Proportional
Reasoning for
Teaching
Mathematics in
Grades 6–8* (Lobato
and Ellis 2010) for
an extended
discussion of
invariance in
proportional
reasoning.

Dougherty and
colleagues (2010)
discuss measurement
in terms of
comparison to a
unit in
*Developing
Essential
Understanding of
Number and Numer-
ation for Teaching
Mathematics in
Prekindergarten–
Grade 2*.

Among many other
things, 2π is the
circumference length
of the unit circle.

We may compare aspects of figures of the same sort—say, two angles or the corresponding sides of two triangles or the areas of two similar shapes. But lurking inside even the most seemingly well-known geometric figure, the square, one can find some surprises. We can ask what is the ratio of the side to the diagonal and receive a mathematically complex answer. Even to speak of *the* ratio masks a presumption about an invariance—namely, that whatever the ratio is for *this* specific square, it will be the same as for any other one. Geometers are often indifferent to the specifics of particular situations; they are far more interested in invariance—by what they discover to be a commonality across the background noise.

For the ancient Greeks, all measurement was explicitly about ratio: the result was not given as a number, but rather as a comparison of one thing with another, such as the sum of the interior angles of a triangle with two right angles—a comparison that reveals a relationship of equality. Although measurement now is often understood as a numerical quantity, it is also in fact a comparison, one involving the unit of measure. When students forget to add the unit at the end of a calculation, they have lost track of this idea of measurement as ratio.

More interesting (and sometimes more complex) mathematics can arise in attempts to compare two *different* things—that is, to compare across categories. One highly significant comparison emerges from comparing curved things with straight things. Whenever we ask about some property of a circle (perhaps the simplest and most regular of curved things), we discover that it usually involves some associated straight component. Consider the example that comes from asking how the diameter compares with the circumference of a circle. How might we make this comparison?

A geometric approach will call attention to the invariance of the relationship between the circumference and the radius, instead of the thing that is to be calculated. In other words, instead of focusing on the formula $C = 2\pi r$, which provides the desired numerical result, a geometric approach features the invariance of the ratio of C to r . The geometric theorem associated with the circumference formula is that the ratio of the circumference of a circle to its radius is invariant, not that this invariant ratio should be interpreted numerically as 2π .

The reason why the ratio is invariant relates to the fact that all circles are similar. The circumference and the diameter are both lengths; under scaling, all lengths will be multiplied by a common scale factor, which means that the ratio of those lengths does not change. Figure 2.1 shows varying quantities of the circumference and the radius accompanied by the invariant ratio (to two decimal places). Because formulas can be so alluring, they can cause us to forget or lose sight of the essential ratio invariance of measurements such as the circumference of a circle.

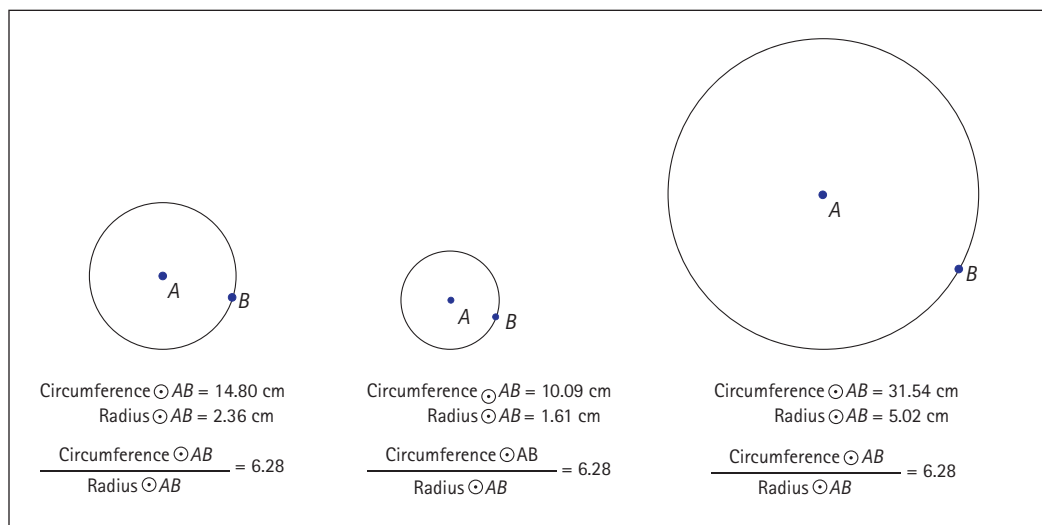


Fig. 2.1. Circumference as an invariant ratio

Behind any formula is an invariance expressed in a theorem. A school task can be to take apart a formula and reconstruct the invariance and express it as a theorem. Unlike the formula, which gives highest importance to the entity you want to calculate, the theorem gives highest importance to the invariant ratio. For example, the ratio of the area of the circle to the square on its radius is invariant. This is due to the similarity of all circles. The ratio of the volume of the cylinder to the product of the square on its radius and its height (the volume of a cuboid) is invariant.

Slope as an embodiment of ratio

So far, we have focused on how ratios are centrally involved in geometry, and particularly in measurement. But geometry can also help in visualizing ratios, which are often inaccessible in a concrete way (for example, we know the two quantities, but their ratio is less tangible). Slope, treated as a geometric notion, provides a visual embodiment of ratio. As a geometric object, slope is not a numerical calculation, but a way to compare two lines. Slope can be treated as an object associated with a certain amount of slant, perhaps depending on the angle that a line makes with a horizontal. In this instance, by focusing on one angle (such as the angle at A in fig. 2.2a), we can treat slope as an example of an invariant across a family of similar triangles. This focus on invariance offers one way of defining slope that does not involve attending to numerical ratio.

Another way of treating slope is to see it as a ratio of two quantities. Similar triangles can offer a direct interpretation of this ratio-driven idea of slope. Triangles ABC , $AB'C'$, $AB''C''$ are similar (see fig. 2.2). The ratio of BC to AC is thus equivalent to the ratio

Big Idea 1



Behind every measurement formula lies a geometric result.

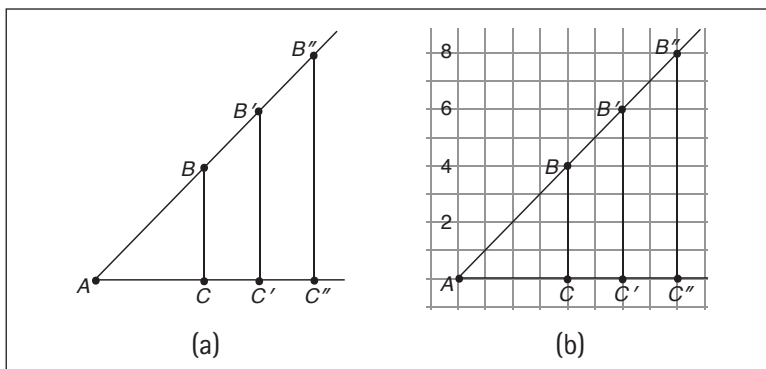


Fig. 2.2. Connecting similar triangles, slope and ratio

of $B'C'$ to AC' and also to the ratio of $B''C''$ to AC'' (and they all are equivalent to the ratio of 1 to 1). In this sense, the slope of the line going through A and B embodies these equivalent ratios.

If we add the Cartesian coordinate system shown in figure 2.2b, we see that slope still embodies the ratios. We also see that it is true that the ratio of the y -coordinate to the x -coordinate is the same for each point on the line (B , B' , B''), as well as for *any* point on that line, including ones that go into the third quadrant (see fig. 2.3). The ratio of dilation for the triangles on the left as images of triangles on the right, with respect to the center of dilation A, would be negative.

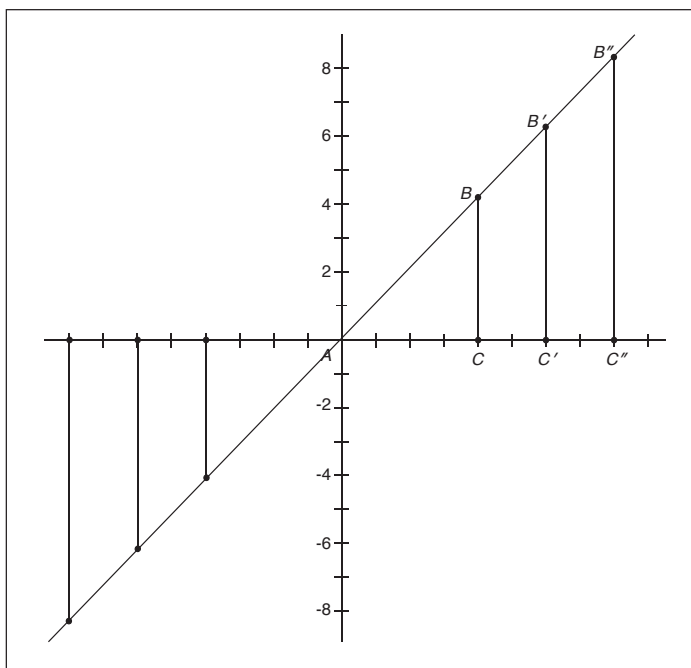


Fig. 2.3. Ratio of the y -coordinate to the x -coordinate remains the same in the third quadrant

Similar triangles that are growing can also provide a concrete way of thinking about rate of change. As the triangle ABC grows, as in figure 2.3, we can ask about the rate of change of one of its sides, of its angles, of its perimeter, and of its area. These are all different rates that can be considered geometrically and visually before attending to numerical quantities. If we imagine moving C to C' continuously (while B moves to B' in the same way), we can see that the angle at A does not change and that the length AC grows at the same rate as the lengths AB and BC —namely, by whatever the ratio of AC' to AC is. What about the rate of change of the perimeter? It also grows at the same rate as each of the lengths—namely, by the ratio of AC' to AC .

Area is a different story, in which considering specific examples might provide better imagery for the rate of growth of area. Consider figure 2.4, in which the length of a side of the triangle has grown by a factor of 2, so segment AC' is twice as long as segment AC . By adding the dotted lines, it is possible to see four copies of the initial triangle ABC in the similar triangle $AB'C'$. The rate of change of the area is therefore not the same as the rate of change of the sides. Tripling the lengths of the triangle will similarly result in producing nine copies of the triangle ABC .

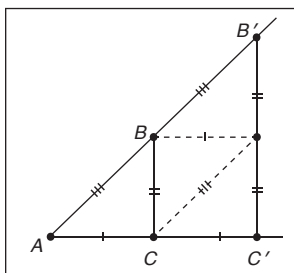


Fig. 2.4. Rate of change of area in similar triangles

Of course, triangles are not the only shapes with which it is possible to provide concrete examples of rate of change. It may be even clearer with squares. Consider figure 2.5, in which dragging the vertex A of the square along the line containing its diagonal effectively produces a family of (similar) squares that share a common vertex. With squares, it may be easier to see the quadratic rate of growth of the area with respect to the side length of the square.

By comparing different geometric objects, then, we can generate different invariant ratios as well as different rates of change. For example, compare the diagonal of the square to its side, the volume of a cube to its edge, or the volume of a cube to its surface area. In addition to providing a firm grounding in the essentially ratio-based idea of measurement, such comparisons of geometric objects can, in turn, provide helpful imagery for concepts such as ratio and rate of change.

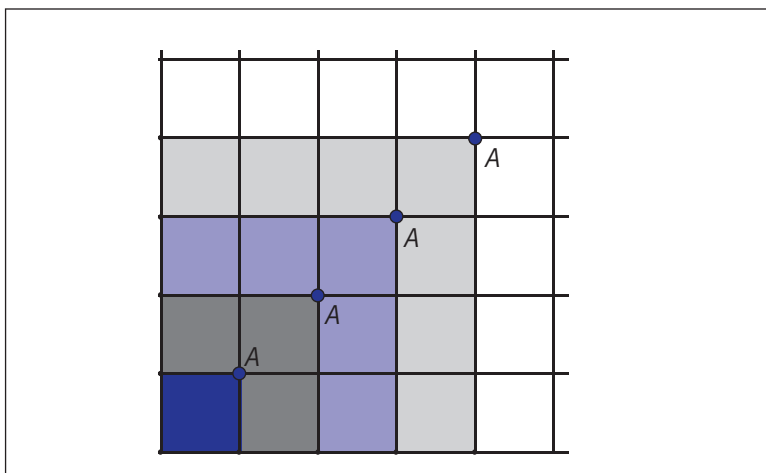


Fig. 2.5. A family of corner-similar squares

Looking Vertically at School Geometry

The big ideas of geometry in this book grow out of and extend naturally to ideas in other grade levels in the geometry curriculum. In this section, we consider both the way in which students encounter shapes in grades 3–5 and the trigonometry ideas that they meet in grades 9–12.

Thinking about shape in grades 3–5

By the time that students begin grade 6, they have a great deal of familiarity with many basic shapes in geometry. They are able to identify and name a range of two- and three-dimensional shapes. The van Hiele model (see, for example, Crowley [1987]) provides a good guide for describing the development of geometric thinking. Teachers using this model are likely to find that beginning sixth graders' thinking about shapes and their strategies for naming and identifying them are at van Hiele level 2—or even, depending on the shape and the context, only at van Hiele level 1. We describe below what we mean by students' "thinking about shapes" and their "strategies for naming and identifying them."

Level 1

At level 1, when children use the word *triangle*, for example, they have in mind a concrete shape that looks like the other things that they have called *triangle* in the past. In this sense, they use the word in the same way that we use proper nouns: to designate a particular, real individual—and we would not call another individual by the same name. However, even if that individual has transformed in a small way, we still use the same name; likewise, children at

this level use the word *triangle* to name not just the most canonical shapes associated with the word (the more or less equilateral triangle with its base parallel to the bottom of the page or the horizon) but also slight transformations of it (perhaps a more isosceles-like triangle). Children at this level identify a triangle by visually comparing the shape with other triangles that they have seen and labeled in this way before. However, if they are shown a shape like any of those in figure 2.6, they will probably not be willing to call it a triangle.

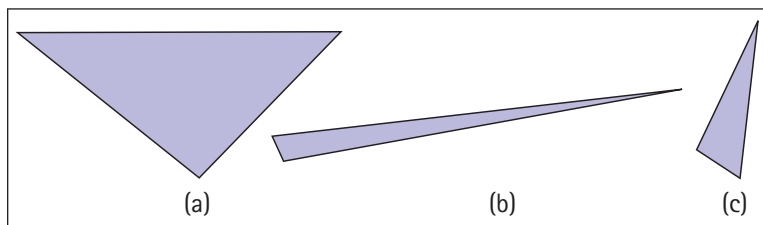


Fig. 2.6. Non-canonical triangles

Level 2

At level 2, and with more experience in geometric discourse, children have encountered several different three-sided shapes, all named *triangle*. They can thus talk about *triangle* as a family of shapes that resemble one another. They are also able to talk about the properties of this family of shapes. For example, they can say that triangles have three sides. As at level 1, they use visual identification, including transformations from one triangle to another, to identify shapes as triangles. They do not use the properties as a means of identifying triangles. In other words, if they are unsure about shapes such as those in figure 2.6, they will not name them triangles because they have three sides. Although properties may sometimes sound like definitions, children at level 2 do not use them in that manner when naming shapes. In fact, a child operating at this level often says that a shape like that in figure 2.6a is a triangle just because it has "three sides and a point." At this level, they might call the shape an "upside-down triangle," but they would certainly not call a square a *rectangle*, for this would be giving a family of shapes that already has a name another one. Different names mean different things. For example, children at this level may designate a particular shape as a square or a diamond but not both, or as a square or a rectangle but not both.

Level 3

At level 3, children make a qualitative shift in their way of thinking about geometric objects. They are able to call two shapes by the same name when they recognize a commonality in verbal descriptions of the shapes. In other words, the criterion for calling a given

shape a triangle is that it fits a verbal description of the definition of *triangle* (e.g., “is a polygon in the plane and has three sides”), and thus they can follow a verbal procedure involving counting the sides—a procedure that will yield the same result no matter which particular triangle they consider. In this sense, children are now working with an abstract object (the verbal description) and not a concrete one (the visual artifact). Of course, this does not mean that their thinking stops relying on immediate visual recognition. They always begin the process of identification with visual recognition, but the verbal procedure implicit in the definition becomes the ultimate criterion for appropriate naming. The idea of a definition comes into play, and the definition is the list of properties considered as both necessary and sufficient. At this level, children can call one and the same shape both a *square* and a *rhombus*, because naming a shape appropriately solely involves checking the conditions specified by the definition.

A central goal of grade 6–8 geometry is to support students in developing a way of talking about and naming shapes that is consistent with van Hiele level 3. This is one reason why our discussion of Big Idea 4 in chapter 1 stressed the important role of naming in geometry and the variety of ways in which names might be used inside and outside of geometric language. It is also why we focused on the importance of developing mental imagery. Before being able to talk about a square being a rhombus, students must have developed a sense of the words *square* and *rhombus* as family names for particular shapes. That is, they must understand these words as names for a range of shapes satisfying shared visual features.

Finally, geometric tools can be indispensable in helping students attend to the properties from which they will eventually be able to build definitions. In particular, the use of construction tools (whether a straightedge and compass or a DGE) demands attention to properties and, in particular, an awareness of the “skeleton” of the shape—that is, the underlying properties that produce it. So, instead of going *from shape to property*, students are encouraged to go *from property to shape*, which is the direction required for naming by definition.



Big Idea 4

Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.

Thinking about trigonometry in grades 9–12

One setting in which triangle similarity is very significant is in the initial definition of the six trigonometric functions: sine, cosine, tangent, cosecant, secant, and cotangent. The very word *trigonometry* suggests the measurement of “trigons,” which we proposed in chapter 1 as a plausible name for polygons with three sides—a name

that would be more consistent than *triangles* with the common naming scheme for most other polygonal shapes.

The sine and cosine of a given angle are usually specified first in terms of ratios of side lengths of a right triangle containing the particular angle. If the angle is itself the starting place (see fig. 2.7), there are an infinite number of right triangles, each of which contains the desired angle. Why does it not matter which one we use?

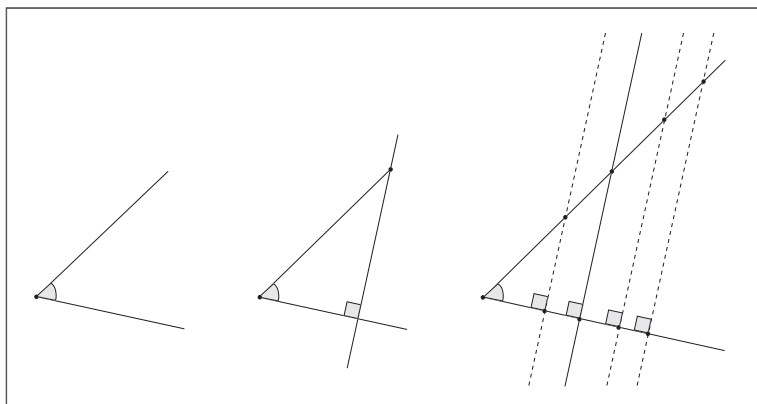


Fig. 2.7. An angle and its associated family of right triangles

Because all of these triangles are similar to one another, the ratio of the lengths of two sides of any one of the triangles is equivalent to the ratio of the lengths of the corresponding sides of any other triangle. Since all six of the trigonometric functions are defined as side-length ratios, it does not matter which right triangle we use. So the whole of trigonometry starts with similar triangles and their properties.

An alternative name for the family of trigonometric functions is the *circular functions*, because relating these functions to circles is a fairly straightforward process. Figure 2.8 shows an important right triangle that is related to the circle and helps to show where two of the six triangular functions' names come from. The Latin verb *tango* (the origin of the name of the Argentinian dance) means "I touch"; a *tangent* in this context is a "touch-line" of the circle. By contrast, a *secant* is a "cut-line" of the circle. As we mentioned earlier, the Latin verb *seco* means "I cut," and it is also the origin of the geometric term *sector* for the region of a circle cut off by a secant line. So in this first right triangle, if θ names the angle in the right triangle at the center of the circle, and if the radius of the circle is 1 unit, then the length of the segment shown on the tangent line is $\tan \theta$, and the distance along the secant line between the center of the circle and the tangent line is $\sec \theta$. Notice that these are definitions by genesis: we are told what to do to find the thing being defined.

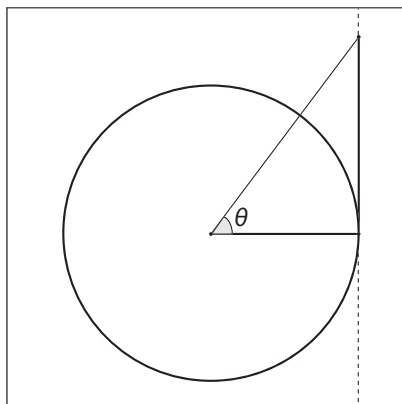


Fig. 2.8. An important right triangle associated with a circle

Inside any circle sits a second right triangle that is similar to the first one (see the smaller triangle in fig. 2.9). This second triangle is the one more commonly drawn in textbooks. *If the radius of the circle is chosen to be 1*, then $\sin \theta$ and $\cos \theta$ are the lengths of the legs of the triangle.

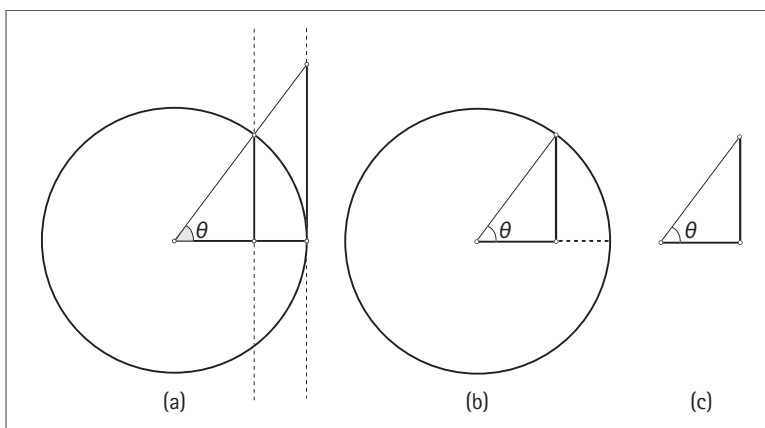


Fig. 2.9. The similar triangles that give rise to the definitions of the trigonometric functions

Applying Pythagoras's theorem to each of these two right triangles in turn produces two of the common trigonometric identities: $1 + (\tan \theta)^2 = (\sec \theta)^2$ and $(\cos \theta)^2 + (\sin \theta)^2 = 1$. Being familiar with similar triangles—and the length ratio properties that are invariant across them—is fundamental to high school trigonometry.

Combining the two perspectives discussed briefly in this section, we see that the unit circle centered at the origin results in $(\cos \theta, \sin \theta)$ as the coordinates of one vertex of the smaller right triangle in figure 2.9a and $(\tan \theta, \sec \theta)$ for one vertex of the right triangle in figure 2.8. Using similar triangles, it is possible to derive the more common algebraic definition of $\tan \theta$ (as the ratio of $\sin \theta$ to $\cos \theta$) as a result.

Conclusion

In both the study of ratio and proportion and the study of trigonometry it can be easy to focus mainly on the numerical or algebraic objects and results involved. One goal of this chapter has been to draw attention to the fundamental geometric nature of both of these arenas of study and to point to ways in which working with geometric images and ideas can help provide concrete meanings and intuitive explanations behind these results. Geometry can be seen as a way of applying numerical and algebraic results, but it can also be thought of as a way of embodying and illustrating them, enriching their meaning.

In our vertical exploration, when we moved down to examine students' understanding in grades 3–5, we mainly concerned ourselves with showing some of the complexity involved in the development in students' thinking about and working with shapes as they make the transition to grades 6–8. This complexity can be all the more difficult to appreciate when teachers and learners are using the same words (e.g., *triangle*, *square*) but are meaning very different things by them. Although we have simply pointed to the complexity in this chapter, our goal in the next chapter is to explore ways in which you can use the ideas developed in chapter 1 to offer your students the support that they need as they make the transition from grades 3–5 and prepare for their next big transition, to grades 9–12.

Challenges: Learning, Teaching, and Assessing

Mathematics (particularly geometry) is shot through with infinity, to paraphrase the mathematics educator Caleb Gattegno (1984, p. 20; see also Beeney et al. 1982). Students need to come to appreciate the fact that geometry is concerned with classes of problems, classes or families of configurations, and classes or families of properties: there are infinitely many quadrilaterals whose diagonals cross, as well as infinitely many quadrilaterals with four right angles. Theorems in geometry deal with invariance across a range of classes, where the range often consists of infinitely many related objects or relationships, like the invariance of the ratio of a radius to a circumference. Seeing geometry from this point of view, you can examine a geometry textbook and other teaching materials (including geometry software) with an eye to designing and selecting tasks that will repeatedly bring students up against invariance. To encounter geometry, students need to encounter invariance over and over and over again.

Few lessons or tasks explicitly enact the big ideas and essential understandings described in chapter 1—and they are not intended to do so. The ideas in chapter 1 describe how you as a teacher, and not necessarily your students, need to understand geometry. Being aware of these ideas might change the way in which you select and modify tasks for use in your classroom, as well as the way in which you talk about the geometric objects under consideration.

In the following section, we present a lesson that is available from the NCTM Illuminations website and intended for use at the middle school level. We propose to examine it in light of the big ideas and related essential understandings that we discussed in chapter 1. Our goal is not to evaluate the task; rather, it is to consider how familiarity with these big ideas might influence the way in which you might work through the task in a classroom setting.

Tessellating with Polygons

The Illuminations lesson “What’s Regular about Tessellations?” is broken up into three parts (see <http://illuminations.nctm.org/LessonDetail.aspx?id=L834>). Its goal is to have students explore regular and semi-regular tessellations. In the discussion of the task that follows, we do not apply the ideas from chapter 1 systemically or sequentially. Instead, we work from the task and its structure and identify places where our big ideas and essential understandings could influence what you would see as important to geometry as a whole.

Regular polygons

The lesson begins with an activity sheet (see fig. 3.1) intended to introduce the word *regular*; it is assumed that students already know the formula for finding the sum of the interior angles of a polygon. We invite you to work through the sheet yourself and think about how the big ideas might manifest themselves in your geometric thinking.

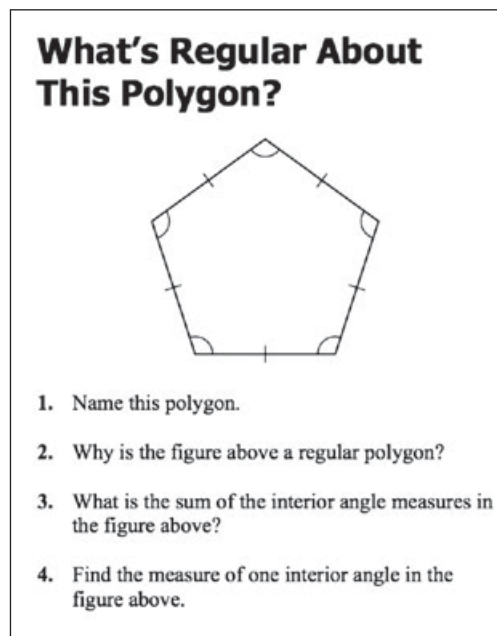


Fig. 3.1. First activity sheet for the Illuminations lesson “What’s Regular about Tessellations?”

As is likely to be evident from your own work on this task, the sequence of questions is meant to lead students to find the measure of an interior angle of a regular pentagon—namely, 108 degrees. We note that no definition of *regular* is given in the activity sheet, but

the diagram is meant to convey some information about what *regular* means (equiangular and equilateral).

Given the focus on visual imagery in Big Idea 2, we might consider how imagery related to the idea of *regular* might be developed. What does *regular* mean in other contexts, and why might that word have been chosen for this one? Are all regular shapes pentagons? Does a regular shape always have to have all sides congruent *and* all angles congruent? What might a three-dimensional regular shape look like? These are questions that not only recognize the significance of naming in geometry, as highlighted in Essential Understanding 4a, but also aim to develop geometric awareness through visualizing, diagramming, and constructing, as in Essential Understanding 2b. With regard to imagery, every regular polygon has rotational symmetry, taking angle to angle and side to side. It is possible to make a family out of the regular polygons and imagine them all situated with a common center (and a common circumcircle—which is the same as saying that every regular polygon is *cyclic*). In doing this, we stress the shared feature of regularity and ignore the different numbers of sides.

The reason that the Illuminations lesson focuses on regular shapes relates to its goal of having students investigate regular and semi-regular tessellations. The basic question that the lesson addresses is, “Which polygons can tessellate the plane?” Looking around, we can usually find some examples quite easily: the square tiles on a floor or perhaps the rectangular panels on a ceiling. It is also quite obvious that circles will not work. But how can we go about identifying others that will? How will we know that we have found them all? One way to answer the general question is to focus on a more restricted, special class of shapes. Regular polygons are one of the most special classes; moreover, we already know that a square (which is a regular quadrilateral) will work. And to make matters even better, we know something about the angles of regular polygons. By keeping in mind the big picture (the general question about polygons), we can remember why it is that we want to focus on regular polygons, but we also work with geometric images to notice properties and invariances that will be useful in the investigation, as described in Essential Understanding 2a.

Although we might already know the formula for finding the sum of the interior angles of a polygon, Big Idea 1 points us to the fact that formulas have associated geometric theorems and that knowing the formula does not mean that we can work geometrically. Given that an interior angle of the pentagon measures 108 degrees and an interior angle of the square measures 90 degrees, we might still find it worth asking whether there are any other angles between 90 and 108 degrees that will produce regular polygons.

Construct a segment in a dynamic geometry environment (DGE). Rotate it around its endpoint by 100 degrees (roughly

Big Idea 2



Geometric thinking involves developing, attending to, and learning how to work with imagery.

Essential Understanding 4a

Naming is not just about nomenclature: it draws attention to properties and objects of geometric interest.

Essential Understanding 2b

Symmetry provides a powerful way of working geometrically.

Essential Understanding 2a

Geometric images provide the content in relation to which properties can be noticed, definitions can be made, and invariances can be discerned.

Big Idea 1



Behind every measurement formula lies a geometric result.

midway between 90 degrees and 108 degrees, though any angle in this range would do). Rotate the new segment around its endpoint by 100 degrees. Do this one more time. You will get a diagram like figure 3.2a, and, if you do it again, like figure 3.2b. For the final segment to join up with the initial one, the angle would have to change. Varying the length of the sides does not change this fact (see fig. 3.2c).

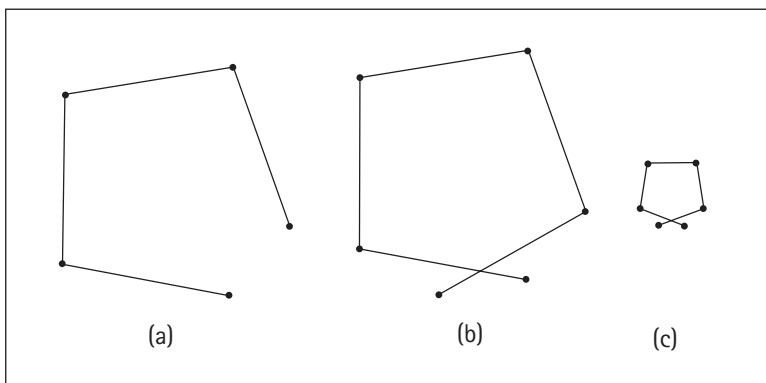


Fig. 3.2. Explorations with a regular pentagon

As with the equilateral triangle (the regular triangle) and the square (the regular quadrilateral), all regular pentagons are similar. This information is useful, because it means that in the investigation of which regular polygons will tessellate the plane, we do not need to worry about the specific length of the sides but only about the angles.

Looking at this more dynamically, we could turn the angle into a parameter. As we vary the angle, we have evidence that there is no value between 90 and 108 for which the final segment produces a regular shape (see fig. 3.3), as well as what might happen if we go past 108 degrees.

We find an indication of continuity here; as we increase the number of sides, we also increase the value of the angle at each side of the regular polygon. The value of the angle was 60 degrees for

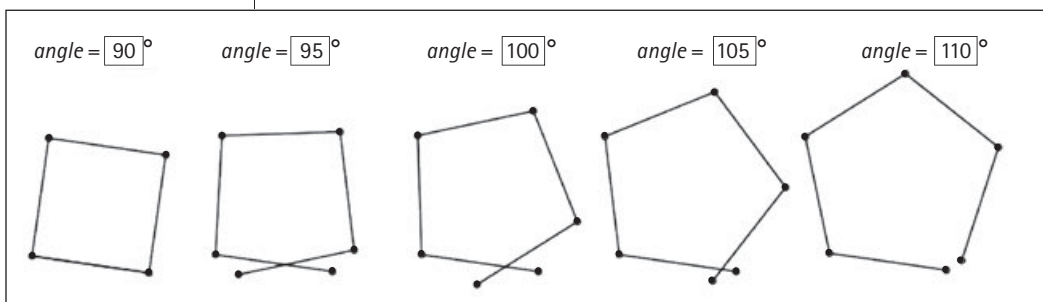


Fig. 3.3. Varying the angles of the regular pentagon

an equilateral triangle, then 90 degrees for a square, and then 108 degrees for a regular pentagon. Once we have identified the angles associated with each consecutive regular polygon, we know that there will not be any regular polygons for any of the angles between them. Furthermore, this observation suggests that there might be a formula giving us the interior angle of a regular polygon when we know the number of sides.

A key geometric idea in moving toward this is *triangulation*, cutting up a polygonal region into triangles. Because of the symmetry mentioned above, the center of the circle is, perhaps, the obvious point to use to connect every vertex of the regular polygon to, creating as many triangles as there are vertices (or sides) in the original. But above all, this book has been about staying with the geometry as long as possible, since it is where a great deal of mathematical meaning, insight, and understanding reside.

With this exploration, we have gone from thinking about the regular polygon as an object to thinking about how it is built, a change that enabled us to relate the value of the interior angle of a regular pentagon to the value of the angles of other regular polygons. As suggested in Essential Understanding 1a, we have been investigating visual ways of seeing how the regular polygons are related to their angle values.

Regular polygons that tessellate

In the next part of the Illuminations lesson, students are asked to create some tessellations by using regular polygons. They have the option of using cutout pieces of paper (many copies of each of the regular polygons) or of using an applet. After exploring with all the different regular polygons provided (up to a dodecagon), the students are asked the following questions (questions 2–4 on the second “What’s Regular about Tessellations?” activity sheet):

2. Which regular polygons will tessellate on their own without any spaces or overlaps?
3. Are there any mathematical reasons why these are the only shapes that will tessellate? (Hint: How many degrees are there in a circle?)
4. Is it possible to tile the plane using only regular octagons? Why or why not?

These questions ask students to observe patterns, to notice mathematical relationships, and to provide reasons and arguments for why specific shapes will or will not tessellate.

Consider Harpreet’s answer to question 2:

I found that the triangle, square, and hexagon worked.

Essential 
Understanding 1a
Decomposing and rearranging provide a geometric way of both seeing that a measurement formula is the right one and seeing why it is the right one.

You might notice that this response refers to one single triangle, square, and hexagon. Knowing the importance of imagery, you might ask Harpreet to visualize two (or more) examples of tessellations with regular triangles. Despite different ways of orienting the triangle, the important idea is that *any* regular triangle works. In particular, changing the size of the triangle does not change the fact that it tessellates, as shown in figure 3.4, which shows triangles on the right that are smaller than those on the left.

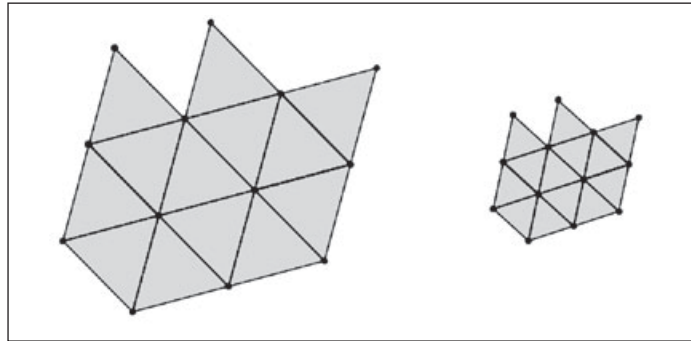


Fig. 3.4. Two tessellations with regular triangles

In light of the importance of naming, you might also ask Harpreet both to be more precise about her choice of names (to use *regular triangle* instead of just *triangle*) and to see whether she can offer names that are different from *triangle*, *square*, and *hexagon*, such as *equilateral triangle* and *regular quadrilateral*. Might it be worth coming up with a special name for the regular hexagon? Doing so might help underline the fact that there exist (infinitely) many hexagons that are not at all regular.

Harpreet gave the following answer to question 4:

You can't tessellate with the regular octagon because it has interior angles of 135 degrees, and 360 is not a multiple of 135.

Harpreet's answer involves a lot of attention to the numerical value of the angles and requires recourse to divisibility. However, a more geometric way of working with this question would involve looking at how the shapes come together in the tessellations that Harpreet tries.

As figure 3.5 shows, a certain number of shapes come together at the vertex in the middle. For a regular triangle, six are arranged around the vertex. For a regular quadrilateral, the number is four; for a regular pentagon, four are too many but three are too few. For the regular hexagon, the number is three. As the number of sides increases, the number of regular polygons coming together at a point decreases. But it cannot possibly decrease to fewer than three; otherwise, there would be only two polygons meeting at a point, which would mean that the angles of that polygon would

have to be 180 degrees—but there is no such polygon. Therefore, it is not necessary to check the polygons where the number of sides is greater than or equal to seven. When question 4 is considered in the more geometric way described, Harpreet's answer becomes much more generalized.

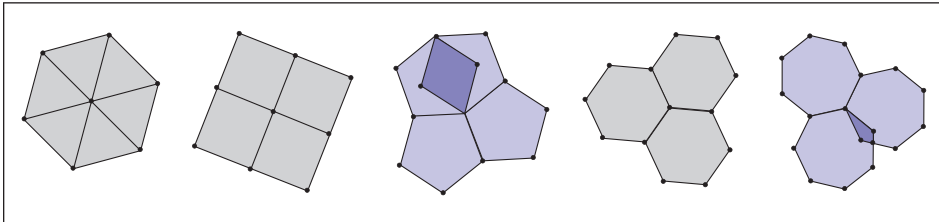


Fig. 3.5. Attempting to tessellate with various shapes

In keeping with Big Idea 1, such an approach also shifts the argument from the formula-driven one that stresses the numerical values of the angle (and requires using facts about divisibility) to working with the transformations of the shapes themselves, noting the continuous qualitative relationship connecting the number of sides to the size of the angles and the number of shapes fitting around a given point.

Big Idea 1



Behind every measurement formula lies a geometric result.

Why particular polygons tessellate

As it is currently posed, question 4 on the second activity sheet for “What’s Regular about Tessellations?” leaves some important questions unanswered: “Why not check a regular 13-gon?” “Why bother checking the decagon when the heptagon, octagon, and nonagon have not worked?” A follow-up question to number 4 asking *why* only three regular polygons tessellate the plane could probe for a more general response and help students move beyond the numerical and empirical realm, as well as allowing them to feel the power of justification.

In addition to seeing whether or not particular shapes tessellate—that is, cover the plane without overlap or spaces (the two core defects that failed attempts at tessellation suffer from)—those investigating a question like question 4 need to give attention to the transformations that relate one shape to its neighbor. Harpreet’s answers focused on the arrangement of regular polygons around a given point, but in working with tessellations, it is important to ascertain whether or not the tessellation could be continued indefinitely in all directions. Looking at the transformations that are used can provide something of a recipe for doing this. For example, with the regular hexagon, one can use translations to generate the tessellation, which will fill space toward the right and below (see fig. 3.6). Reversing the vector of translation will fill the space to the left and

up. For the regular triangle, it can be more convenient to use reflection and rotation.

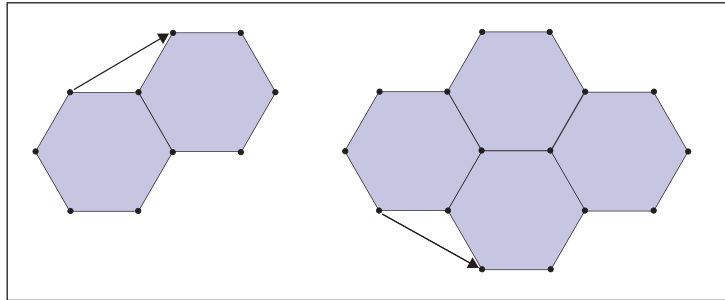


Fig. 3.6. Generating the tessellation through repeated translations

Later, the Illuminations activity asks the students to define *regular polygon* and *regular tessellation*. Because the activity unfolds in this fashion, the making of the definition comes after the exploration. This enables students to connect the different examples that they have worked with to the new terms that they are defining, as suggested in Big Idea 4. Since for geometric objects, the properties that they do not have are as important as the ones they do, this would be a good opportunity to evoke non-examples by posing such questions as, “What is an example of a polygon that is not regular?” “In what ways can a polygon fail to be regular, and how do these ways interact (if they do)?” and, “What is an example of a non-regular tessellation?” (A teacher who follows the lesson through to semi-regular tessellations, in which two regular polygons are used in combination, might suggest ways of extending this line of investigation by posing such questions as, “Is it possible to combine three regular polygons to tessellate the plane?” and “What about four?”)

Given what our investigation has shown about regular tessellations, can we explain why rectangles or parallelograms can tessellate the plane? Or even less regular quadrilaterals than these? Students are often able to imagine what these tessellations would look like. And they are usually very quick to assert that although these special quadrilaterals can tessellate the plane, an arbitrary quadrilateral could not. They often say the same thing about arbitrary triangles.

Students can explore both of these latter general cases by using different types of tools, including cutout paper shapes or DGEs. Such explorations can lead students to notice that their arbitrary triangles or quadrilaterals need to be rotated to fit together. Figure 3.7a shows six copies of the same triangle, with each one as the rotation image of the one beside it around the midpoint of the side that they share. Using the fact that the sum of the angles of a



Big Idea 4

Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.

triangle is always 180 degrees, we can see that any three adjacent triangles will meet at a vertex to create a 180-degree angle also—and thus, two pairs of three triangles will produce the 360-degree angle that is required to fill the space around the vertex. Figure 3.7b shows how the four angles of a quadrilateral meet at a vertex. Each of these tessellations can be extended by continued rotation about the midpoint of a side.

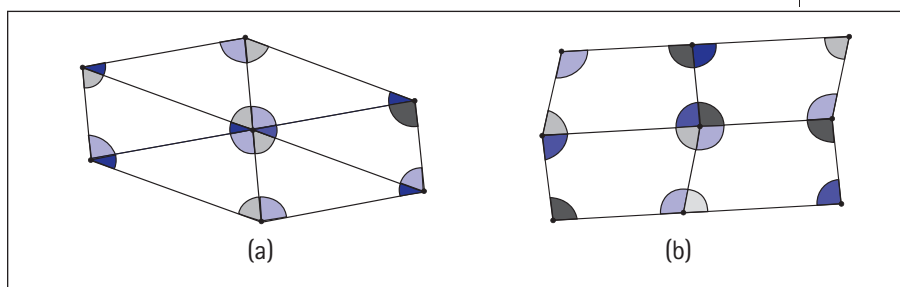


Fig. 3.7. Tessellations of an arbitrary triangle and an arbitrary quadrilateral

Use of Tools in the Lesson

We briefly mentioned the tools that the Illuminations lesson suggests as ways of working with regular tessellations—namely, either paper or the applet provided at the Illuminations site. Certainly, the applet offers a degree of precision and speed that would be lacking with paper. Moreover, students working with paper would be likely to give much less attention to the way in which shapes are rotated and translated (and maybe even reflected) to fit together. The applet, by contrast, allows students to make the translations by dragging and to carry out the rotations even more explicitly, by using the applet's rotation tool. The applet also enables the students to zoom in and out, which might help them become aware of the fact that the actual size of the polygons is unimportant. Finally, the copying tool might help the students think of how the tessellation can be seen as units that are repeated. The applet might not just be faster and more precise but encourage mathematical actions that the paper polygons cannot. Recall that Essential Understanding 3a directs attention to thinking about how tools not only provide new sources of imagery for geometric thinking but also mediate geometric objects and processes in different ways.

Carrying this line of thinking further, we notice that neither of these tools enables students to think explicitly about rotations, reflections, and translations—three of the four types of isometry. However, if the students were to carry out the exploration by using hinged mirrors, they would find that they could use reflection to generate all the tessellations. They would be starting to turn the

Essential Understanding 3a

Tools provide new sources of imagery as well as specific ways of thinking about geometric objects and processes.

process of reflecting individual shapes into an object—the tessellation. Moreover, this process would be explicitly geometrical, in contrast to the process involving either the paper polygons or the applet.

In a dynamic geometry environment, students could also use translation and rotation, in addition to reflection, to generate their tessellations. These transformations could then be used to explore whether other shapes also tessellate. In particular, a DGE would enable students to drag their tessellations of arbitrary triangles or quadrilaterals to test different methods of constructions across a range of shapes. Instead of being a lesson that develops and extends students' understanding of the sum of the interior angles formula, this lesson could be one that focuses on applying transformations and using symmetries to analyze mathematical situations. Indeed, the choice of tool could greatly shift the focus of the lesson, in addition to changing the way in which students might think about the tessellations they have created.

Thinking about Assessment

The main result of the Illuminations lesson is a theorem asserting that the only regular polygons that tessellate the plane are the regular triangle, the regular quadrilateral, and the regular hexagon. Similarity allows us to speak of “*the* regular _____” (for example, “*the* regular hexagon”), without attention to different side lengths. One goal of assessment is not so much to see whether the students remember which regular polygons work but to see whether students have developed strong imagery related to the new concepts *regular polygon* and *regular tessellation* and whether they can articulate the reasoning behind their assertions.

Assessment might also involve producing drawings of polygons or tessellations, prompted by questions such as the following:

1. Can you make a polygon that has congruent angles but not congruent sides? What name would you give it?

Many students would offer the rectangle as a solution to this problem, since it is one of the few equiangular shapes they know of that has a name. Tools such as a protractor or a DGE might help students construct new shapes that are equiangular but without angles that are 90 degrees. For a quadrilateral, the rectangle is the only possibility, since each angle must be 90 degrees, given that the sum of the angles is 360 degrees and four angles are needed. Similarly, the angles of the pentagon will have to be 108 degrees. A family of pentagons can be made that might be called the “equiangular pentagons” (see fig. 3.8), in which the regular pentagon is one special case (and the only one that does not satisfy all the conditions of the question).

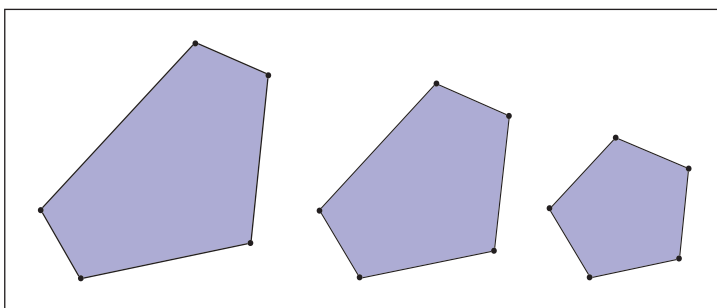


Fig. 3.8. Equiangular pentagons (all angles are 108 degrees)

2. Draw three different tessellations that are not regular tessellations.

In responding to this question, students again might find it easier to generate examples such as the rectangle, the parallelogram, and the rhombus. It could be helpful to challenge students to think of different types of polygons, such as those that are not convex or those that do not have reflectional symmetry. Decomposition could be one way of creating new shapes, as could rearrangement. Having to name these new shapes would help students describe the particular properties that they have, as well as find classes of shapes that tessellate and have the same properties. So, for example, the shaded polygons in figure 3.9 are all ones that tessellate, since they can be rotated around the purple point to create a rectangle (which we know tessellates). The triangle and the trapezoid may seem quite simple, but, as the hexagon and the dodecagon illustrate, a wide variety of unusual shapes could be created this way. These might be called the “half-rectangle polygons” to draw attention to the fact that they are all half the area of a rectangle. Alternatively, they might be called the “rotatable-into-a-rectangle polygons” to draw attention to how they can be tessellated.

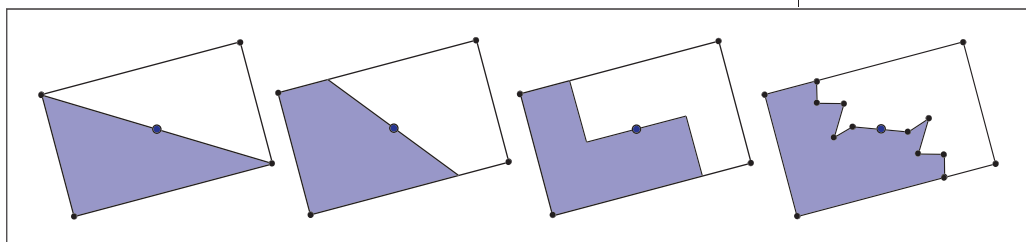


Fig. 3.9. A new family of tessellating polygons

Further questions might involve asking students to use their own words to explain why the theorem that regular hexagons, squares, and equilateral triangles are the only regular polygons to tessellate the plane holds. Or students might be asked why a similar theorem—namely, that *any* triangle or quadrilateral tessellates—does not hold for an arbitrary pentagon. They might also be challenged to

find particular cases of pentagons that do tile the plane, such as the “house-shaped” pentagon, composed of a square with an equilateral triangle with the same side length on top. The “house pentagon” exploration would require that students generate images of pentagons that are not regular and attend to the importance of the angle measure of any regular pentagon. Indeed, consider the non-convex example shown in figure 3.10. The building of this non-convex pentagon began with an equilateral triangle (shown in fig. 3.10a), which is known to tessellate the plane. Then a triangle on the right side was cut off and rotated by 60 degrees to the left side. This new non-convex pentagon was then rotated around the top vertex of the triangle by 60 degrees. Because every piece that was added fits exactly into the place of a corresponding piece that was cut off, rotating the original piece five times—thereby obtaining six pieces in total—produces a regular hexagon, a figure that is known to tessellate the plane.

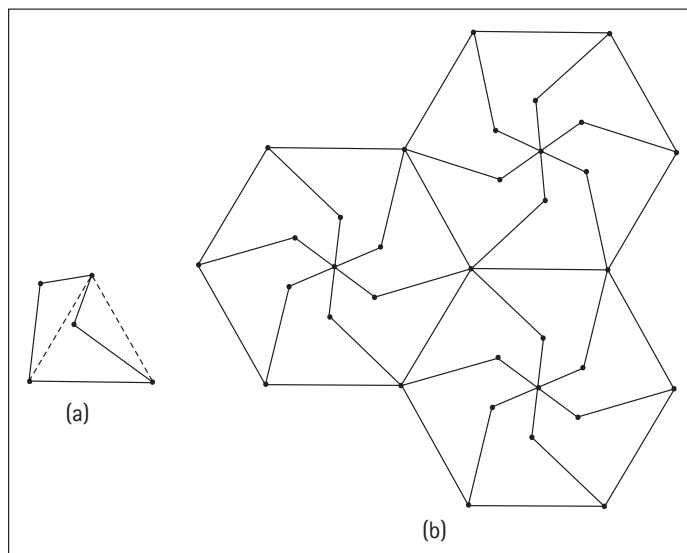


Fig. 3.10. A non-convex pentagon that tiles the plane

A given regular polygon that tessellates can be decomposed and rearranged in infinitely many ways to create a tessellating pentagon. However, there are also pentagons that tessellate and are not combinations of or decompositions of regular polygons. In fact, fourteen such pentagons are now known, but nobody has been able to prove that they are the only ones (Wells 1991). Figure 3.11 shows two examples of the same class of pentagonal tiles (two pairs of adjacent sides are congruent and the angle between these sides is right). In each case, the purple outline in the figure shows how a unit composed of one such pentagon and three images resulting from rotating it around the right angle can be simply translated to continue the tessellation.

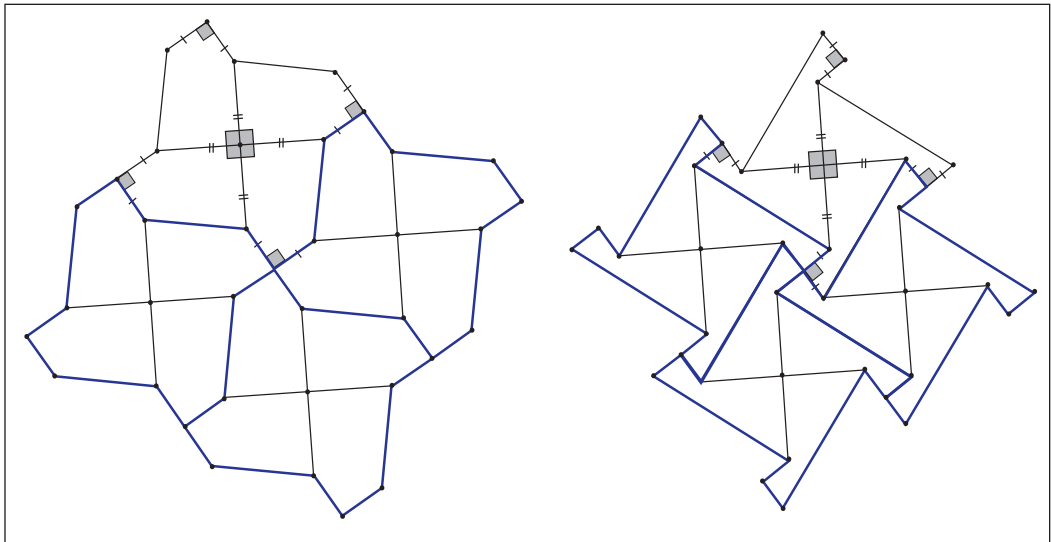


Fig. 3.11. Tessellations arising from two pentagons that satisfy the same properties

Both in these assessment questions and in the elaboration of the discussion of the Illuminations activity, we have tried to exemplify the way in which your awareness of the big ideas and their associated essential understandings might alter the focus of your students' geometric work. Of course, not all geometric tasks are equally amenable to each of the big ideas. In fact, even in our case study, some were much more prevalent than others. Big Idea 1 is very focused on measurement and on increasing the geometric aspect of measurement, so it will not necessarily be relevant to tasks and lessons that are more purely geometric in nature (or where there is no obviously quantifiable variable). It also offers an entirely new way of thinking about measurement, so lessons that relate to measurement would change much more significantly in response to this big idea than to others. Big Idea 3 draws attention to the role of tools in geometric thinking; its main impact on teaching might be to help you become more aware of how the tools that you are offering are affecting the way in which your students think and thus encourage you to try some new tools or to use a range of tools to broaden your students' understanding.

Conclusion

We see Big Idea 2 and Big Idea 4 as the ideas that overarch most school geometry. Yet, these ideas are also very subtle in their influence and scope: they stand to affect all your geometry work in the classroom. Big Idea 2 emphasizes the role of visual imagery in geometric thinking and draws attention to the way that working with

Big Idea 1



Behind every measurement formula lies a geometric result.

Big Idea 3



A geometric object is a mental object that, when constructed, carries with it traces of the tool or tools by which it was constructed.

**Big Idea 2**

Geometric thinking involves developing, attending, to and learning how to work with imagery.

**Big Idea 4**

Classifying, naming, defining, posing, conjecturing, and justifying are codependent activities in geometric investigation.

**Big Idea 1**

Behind every measurement formula lies a geometric result.

**Big Idea 3**

A geometric object is a mental object that, when constructed, carries with it traces of the tool or tools by which it was constructed.

imagery can help students develop more powerful ways of thinking about properties, definitions, conjectures, and theorems. When you are working numerically in geometry, Big Idea 2, together with Big Idea 1, encourages you to consider how you might exploit visual aspects and, in particular, draw on visual thinking that is dynamic. Recognizing the importance of vocabulary and classification in geometry, Big Idea 4 gives you permission, as it were, to spend time doing archeological work with new words. This does not necessarily mean doing the kind of etymological exploration that we did in chapter 1, but it does mean focusing on the names themselves as possible sources of meaning and as important things that have been developed through historical and mathematical processes.

Finally, Big Idea 3 highlights the way in which tools can become a visible and central part of working geometrically. Inviting students to solve a given problem by using different tools (say, a geoboard and a DGE) is tantamount to inviting them to solve a problem by using a different method. Each tool will highlight and require attention to different properties or relationships, and will thus involve different methods. In particular, we have emphasized the value of dynamic geometry tools in helping students attend to invariance and generality.

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